



Create Custom Steps with SAS® Studio

Learning Lab

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Data Engineering and Decisioning
SAS Education Content Development Group

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2026

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Learning Lab

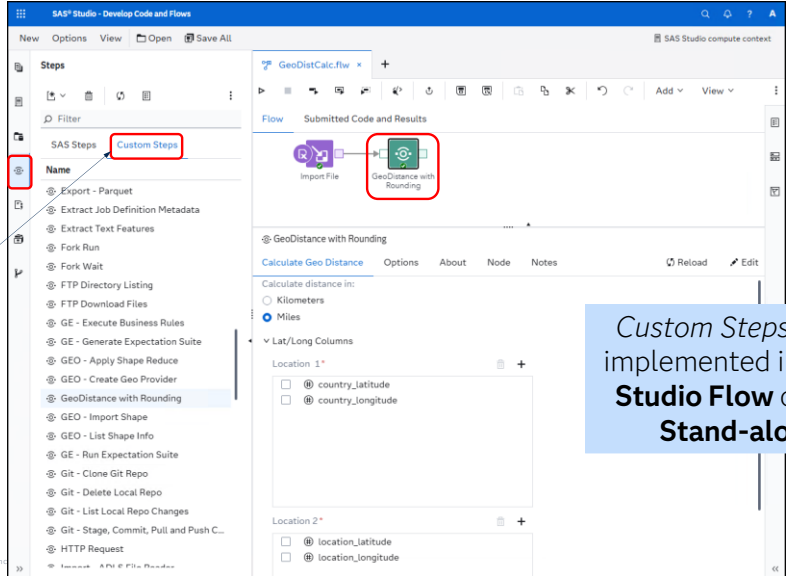
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What is a Custom Step?



The screenshot shows the SAS Studio interface. On the left, the 'Steps' pane is open, displaying a list of available custom steps under the 'Custom Steps' tab. A red box highlights the 'Custom Steps' tab, and another red box highlights the 'GeoDistance with Rounding' step in the list. A callout box points to this step with the text 'Listing of available Custom Steps'. In the center, a flow diagram shows an 'Import File' step connected to a 'GeoDistance with Rounding' step, which is also highlighted with a red box. On the right, the 'Submitted Code and Results' pane shows the configuration for the 'GeoDistance with Rounding' step, including options to calculate distance in miles or kilometers and select location columns. A callout box points to the flow diagram with the text 'Custom Steps can be implemented in a SAS Studio Flow or used Stand-alone.'.

Listing of available Custom Steps

Custom Steps can be implemented in a **SAS Studio Flow** or used **Stand-alone**.

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Custom Steps enable you to create a user interface to complete a specific task. Available custom steps are listed on the **Steps** pane in **SAS Studio** on the **Custom Steps** tab.

They can then be used in multiple SAS Studio Flows or stand-alone to ensure a common, repeatable process.

Designing a Custom Step

The screenshot displays the SAS Studio Analyst interface for designing a custom step. The interface is divided into several sections:

- Control library for adding pages and various controls:** A panel on the left containing a list of controls such as Data, Options, About, Common, Check Box, Color Picker, Date and Time Picker, Drop-down List, File or Folder Selection, List, Link, Numeric Stepper, Option Table, Radio Button Group, Section, Text, Text Area, Text or Numeric Field, and Column Selector.
- Pages:** A tabbed interface at the top with tabs for Design, JSON, and Program. The Design tab is currently active.
- Page Design Canvas:** The central workspace where the user interface is designed. It shows a "Select input table:" field and a "Select columns to print:" field with an "Add columns" button.
- Properties: Column Selector:** A panel on the right showing the properties for the selected control. It includes fields for ID (varList), Label, and options for making the control read-only, hiding it at runtime, and determining where values come from (Table or Column selector).

Annotations in the image include:

- A red box around the "Design" tab and the "Control Library" panel.
- A red box around the "Page Design Canvas" and the "Properties: Column Selector" panel.
- A red box around the "ID:" field in the Properties panel.
- A red box around the "Select input table:" field in the Page Design Canvas.

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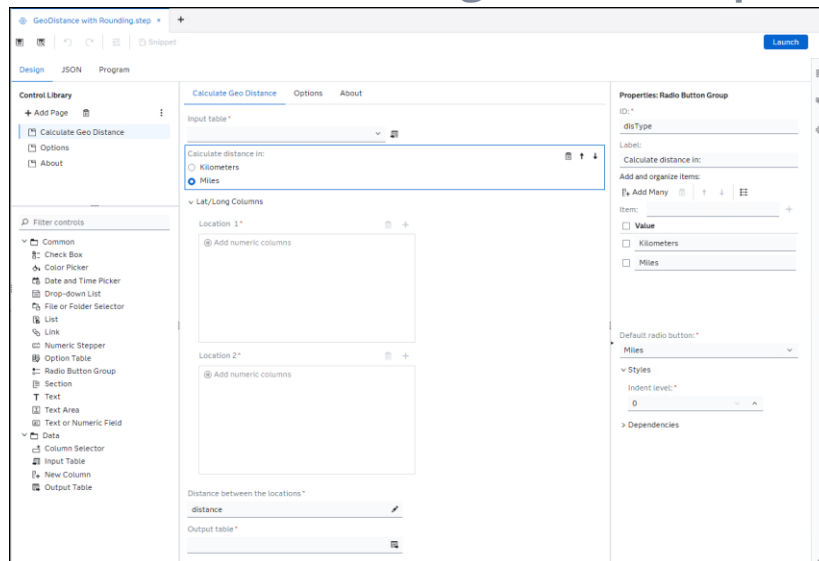
The **Design** tab is a visual interface to help you quickly build the JSON for the user interface for your custom step. This JSON is visible on the **JSON** tab. A **SAS Studio Analyst** license is required to use the **Design** tab; if you just have a **SAS Studio Basic** license, then you will need to design your custom steps using only the **JSON** tab.

On the left-hand side is the Control library for adding **pages** and **controls** to the user interface.

Pages become tabs in the user interface for the custom step. Each page has its own page design that you customize by adding controls to the **Page Design Canvas**.

For each page or control their **properties** are set on the right-hand side. Each control must have a unique ID within the Custom Step.

GeoDistance with Rounding Custom Step



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This is the custom step you will build in the hands-on exercise.

Create Custom Steps with SAS® Studio

Hands-on Exercise

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[Create-Custom-Steps-with-SAS-Studio/Create Custom Step.md at main · SASInnovate2026/Create-Custom-Steps-with-SAS-Studio](https://github.com/SASInnovate2026/Create-Custom-Steps-with-SAS-Studio)
<https://github.com/SASInnovate2026/Create-Custom-Steps-with-SAS-Studio>

Create Custom Steps with SAS Studio

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Exercise Description

In this hands-on exercise you create a custom step that calculates the geo-distance between two latitude and longitude locations in either kilometers or miles and provide the option to round the result. After the custom step is built, you test it in stand-alone mode and then use it in a SAS Studio Flow.

Log in to SAS Viya

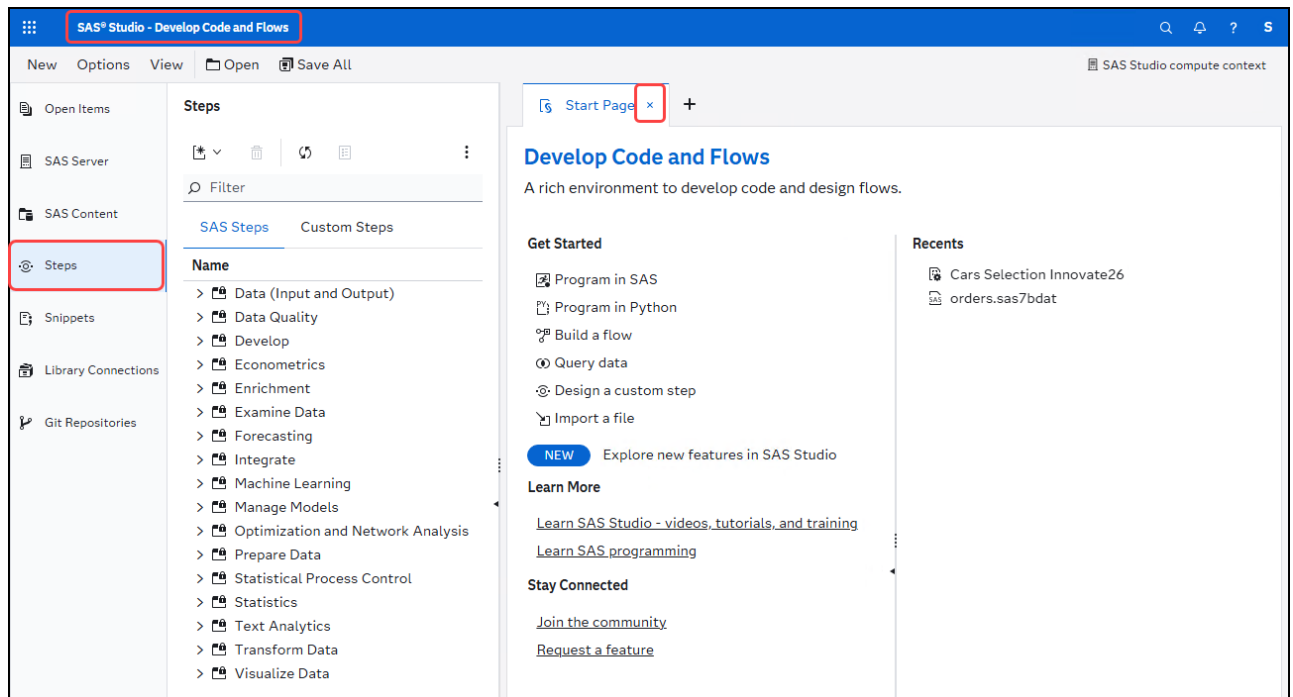
Open a new window in the *Google Chrome* browser and select the **SASLanding** bookmark.

- ID: **student**
- Password: **Metadata0**

Select **No** when prompted about accepting *Admin* privileges.

Create a Custom Step

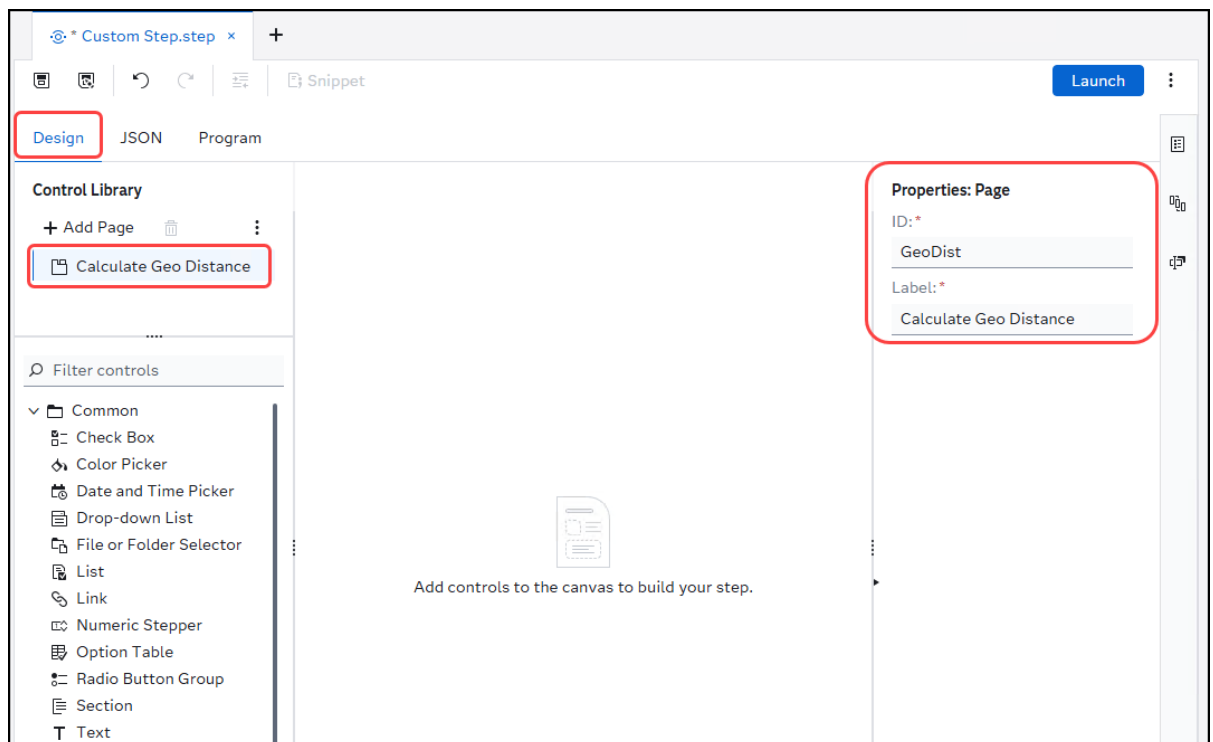
1. Select  → **Develop Code and Flows** to open *SAS Studio*.
2. Click **x** to close the Start Page, if it is open. Also, close any other open tabs, if needed.
3. Select  to view the **Steps** pane.



4. Select  → **Custom step quick start** to create a new custom step.

5. On the **Properties** for the *Page* on the *Design* tab, enter the following:

- ID: **GeoDist**
- Label: **Calculate Geo Distance**



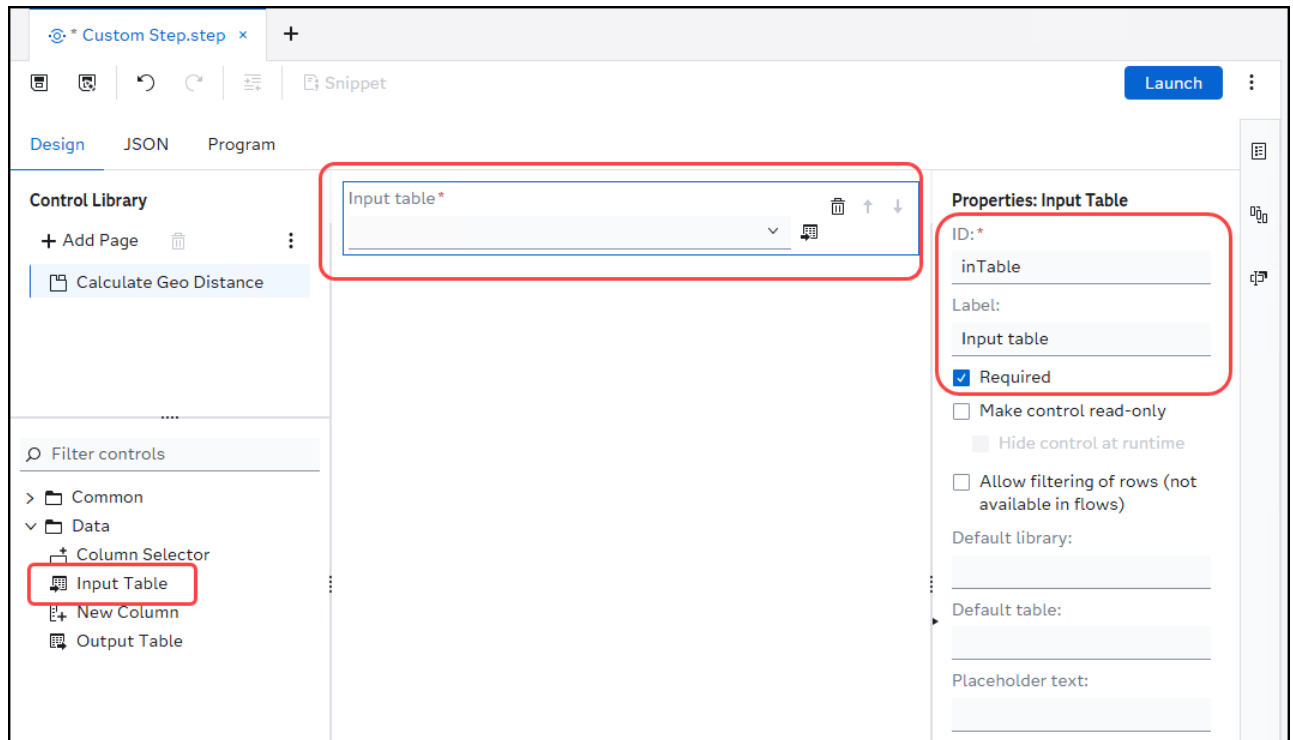
6. Drag the **Input Table Data** control onto the canvas.

7. Enter the **Properties** for the *Input Table*, enter the following:

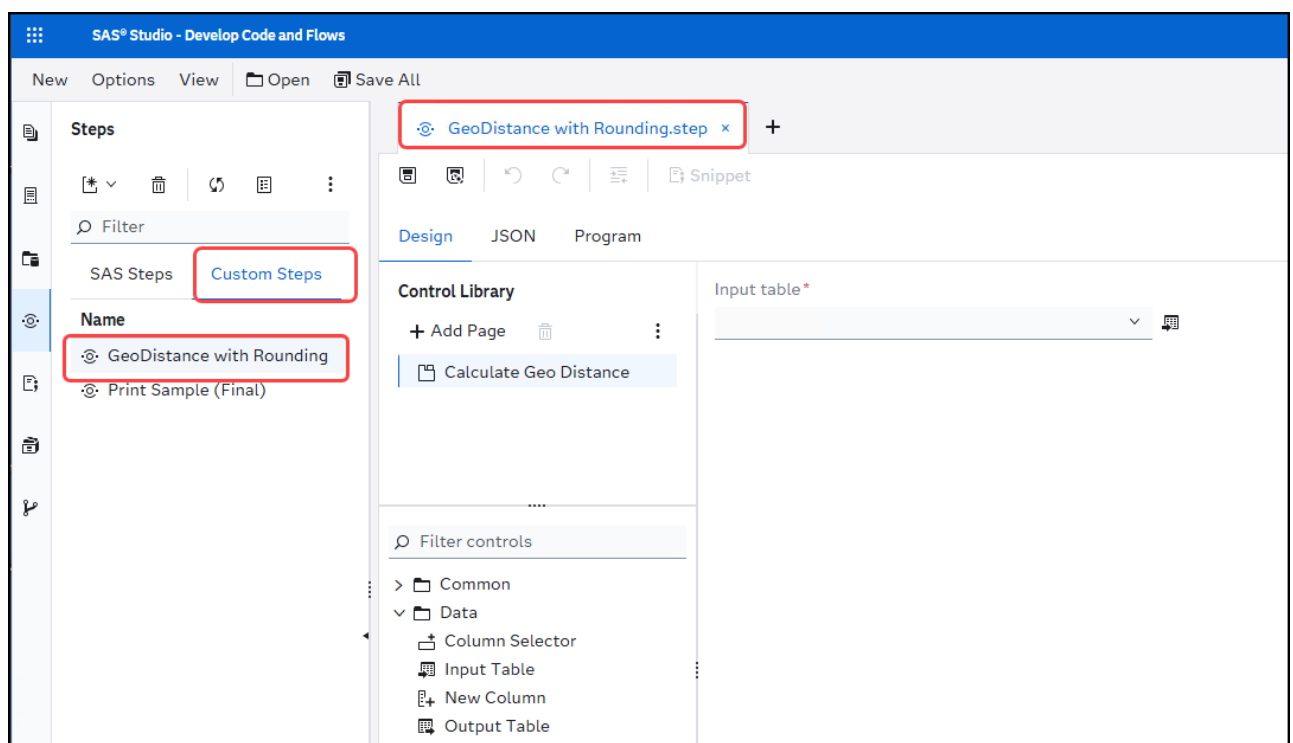
- ID: **inTable**
- Label: **Input table**

- ☒ Required

!! The **ID** names for the controls are used in the code for the custom step, so ensure you enter them correctly; otherwise, you will either need to edit the code or the control ID.

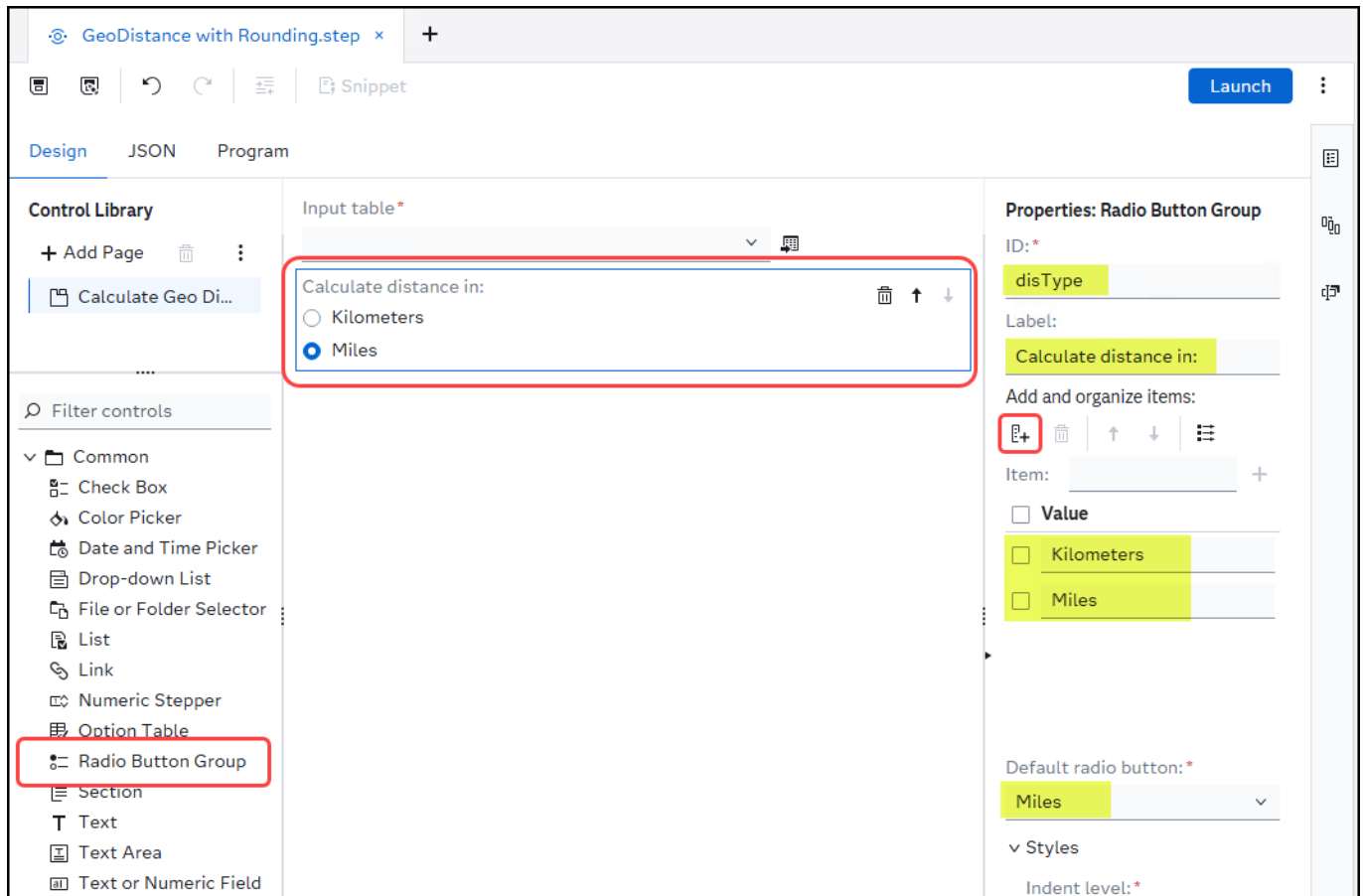


8. Select  to save the custom step.
9. Navigate to **SAS Content** → **Public**.
10. Enter **GeoDistance with Rounding** for the *name* and click **Save**.
11. Select the **Custom Steps** tab and confirm the saved step is listed there.

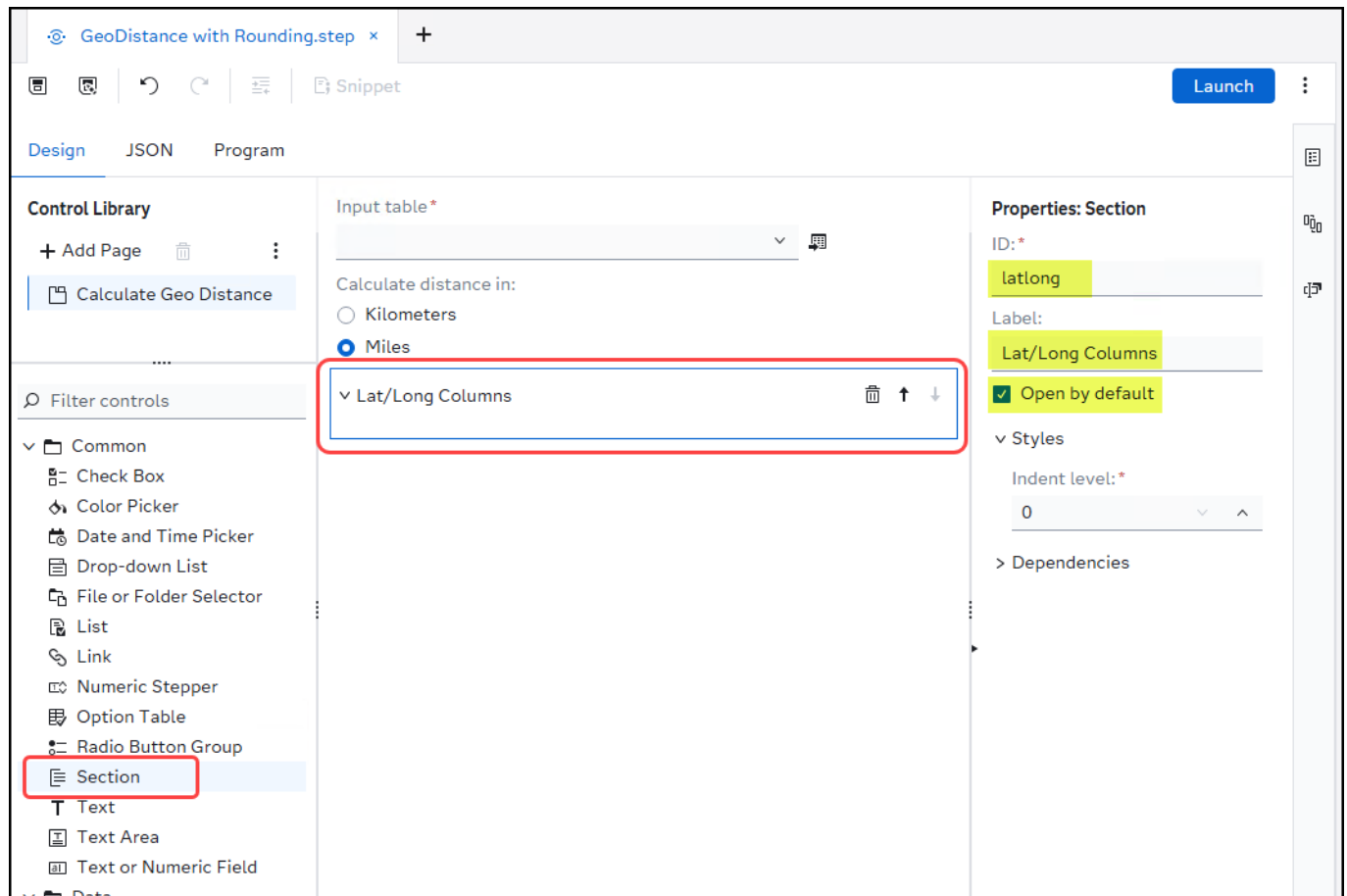


Add a **Radio Button Group** *Common* control to the canvas. Enter the properties as depicted below.

Click  to add an item to the radio button list.



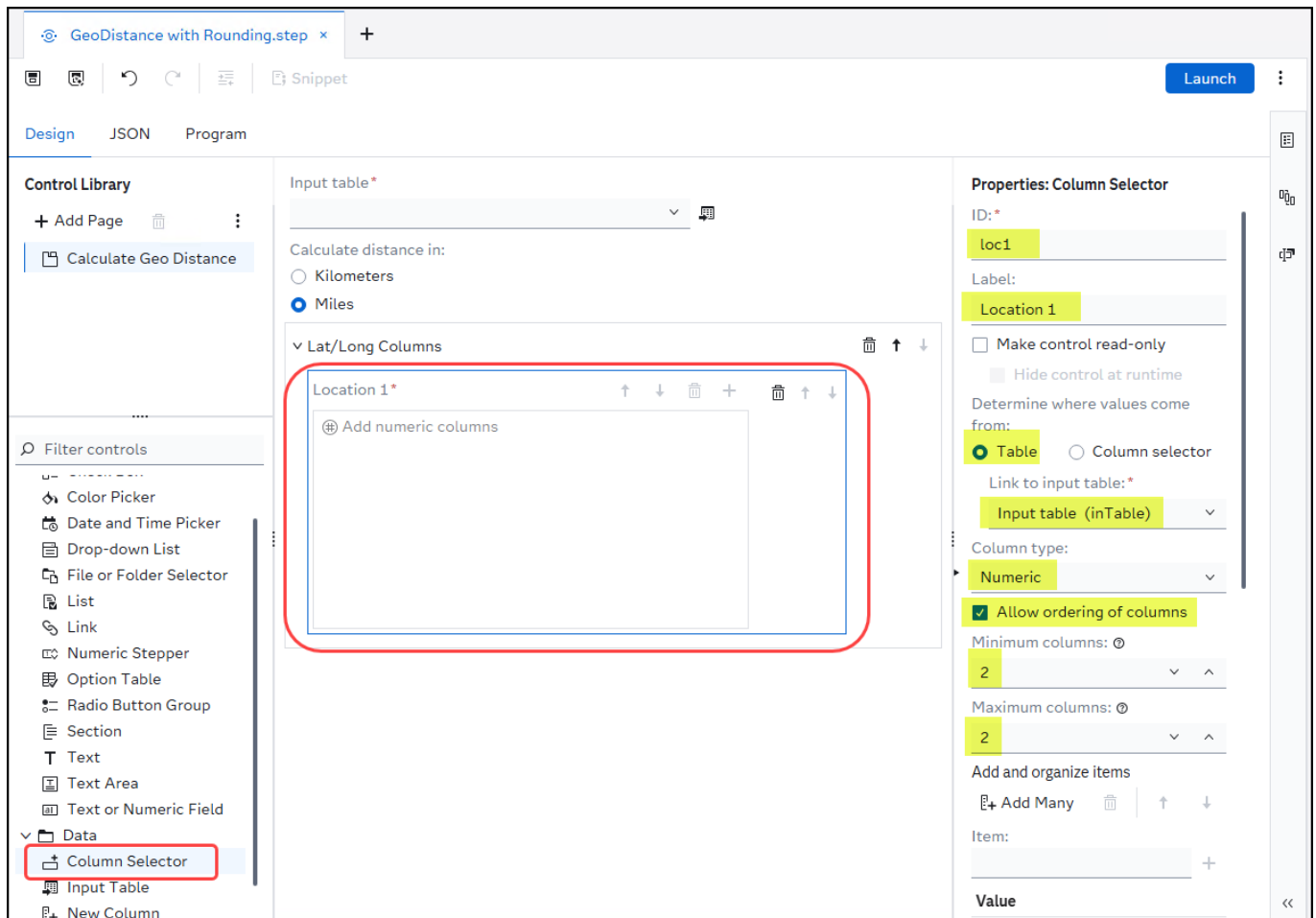
Add a **Section Common** control to the canvas. Enter the properties as depicted below.



Save the changes to the custom step.

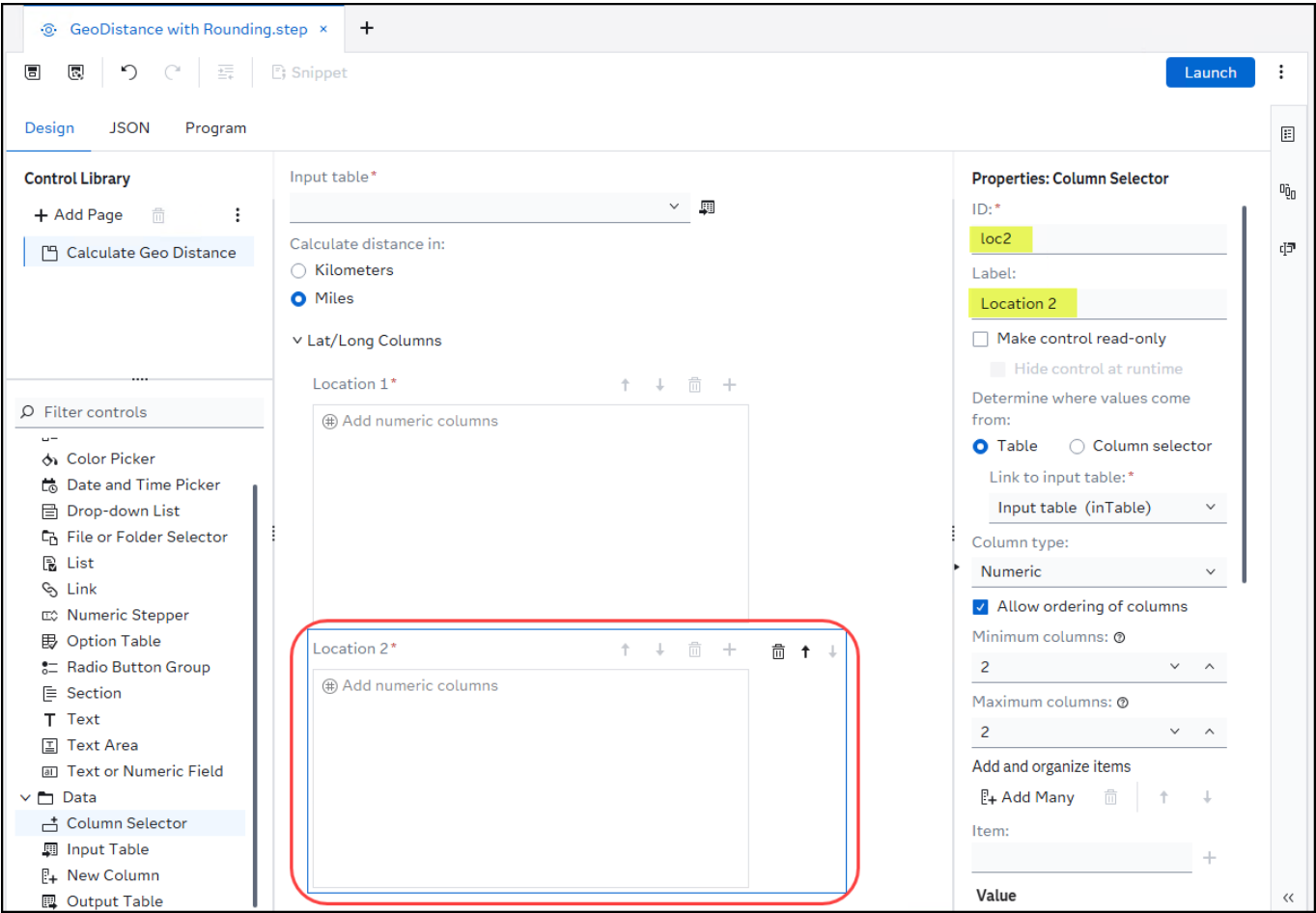
Add a **Column Selector** *Data* control to the canvas in the *Lat/Long Columns* section group. Enter the properties as depicted below.

- Right-click the control once added to the canvas and select **Move to section → Lat/Long Columns** to place it in the section.



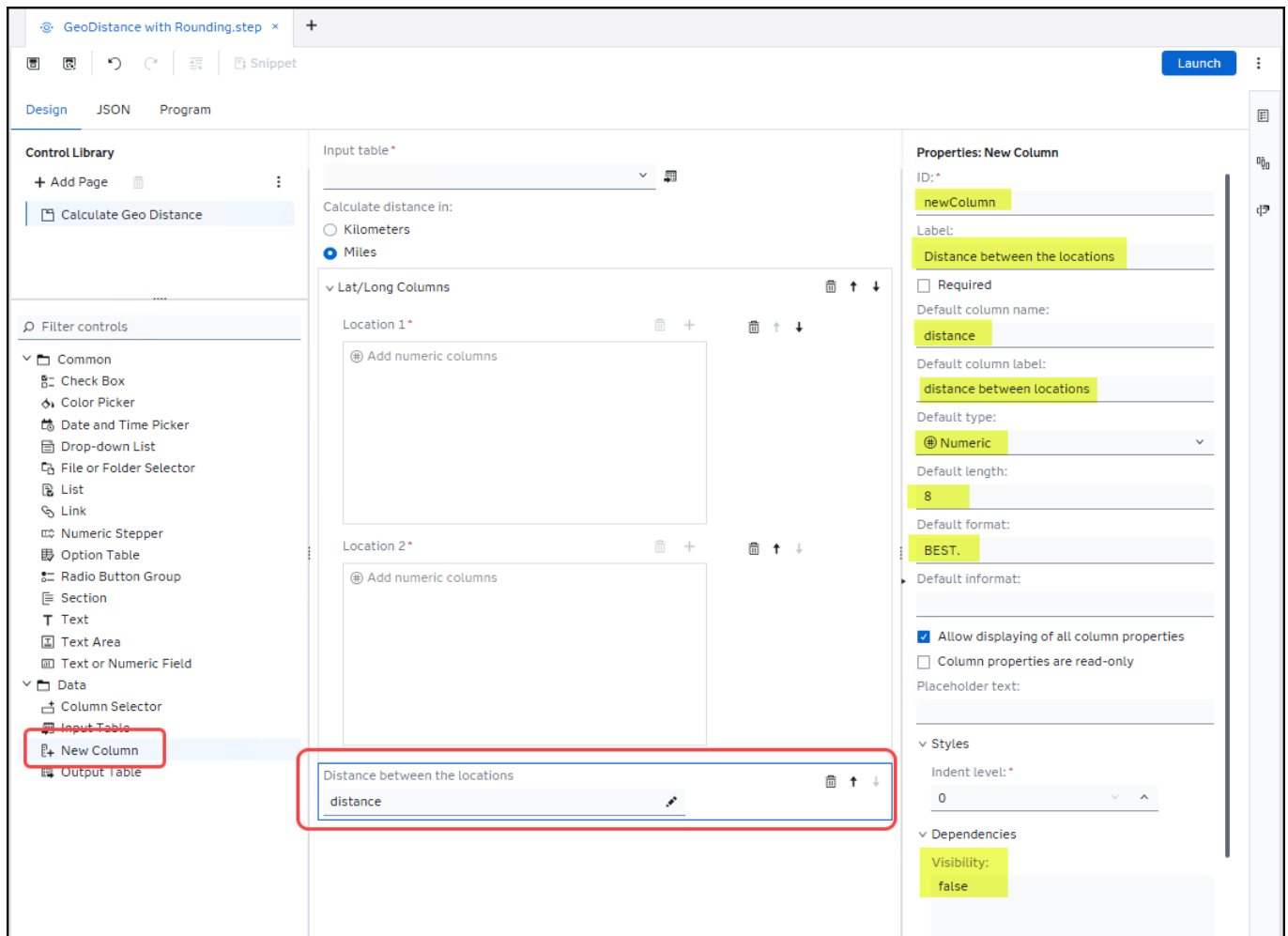
Duplicate the *Location1 Column Selector* control and paste it below in the *Lat/Long Columns* section group. Edit the properties as depicted below.

Right-click the *Location 1* control once added to the canvas and select **Duplicate**.



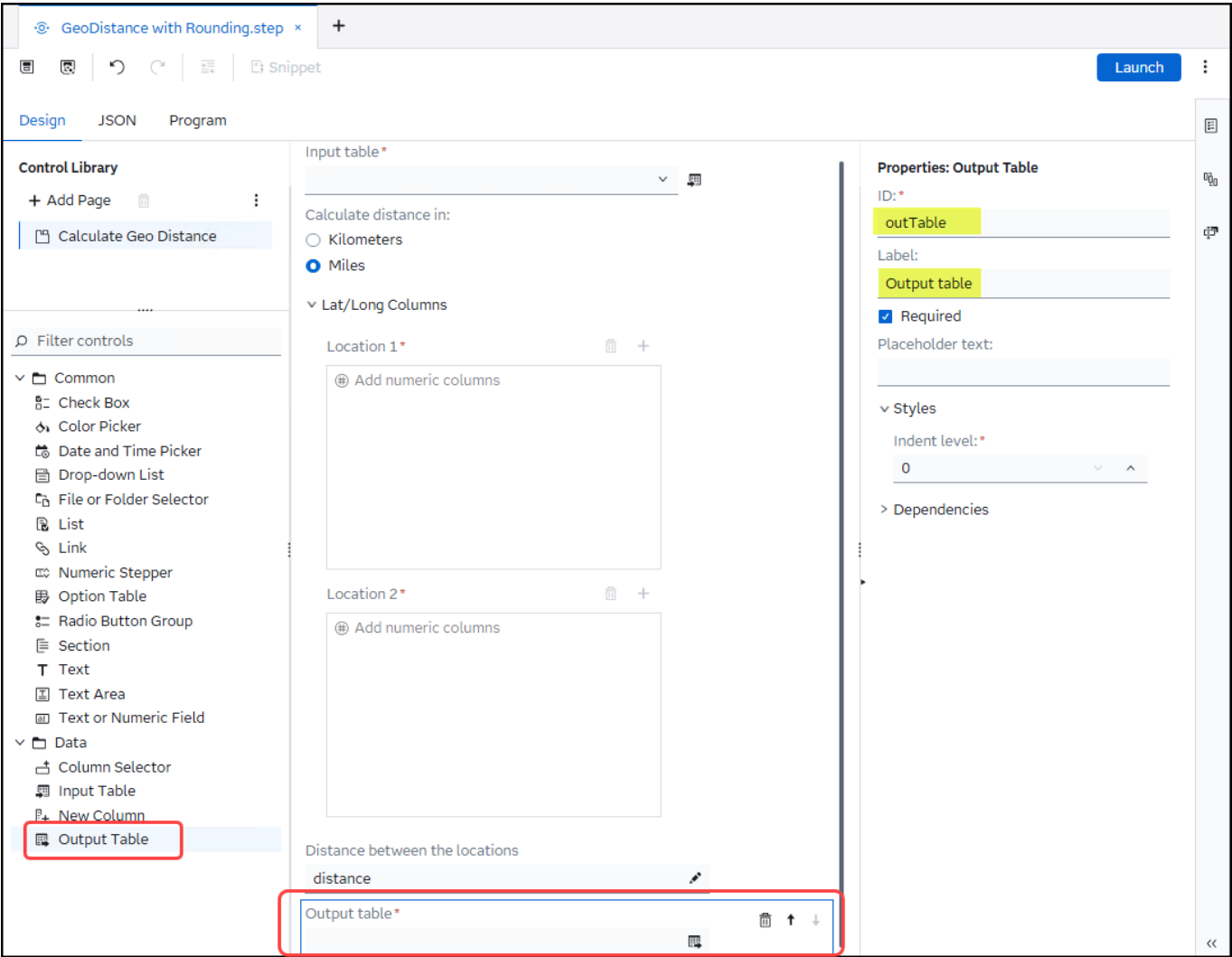
Save the changes to the custom step.

Add a **New Column Data** control to the canvas. Enter the properties as depicted below.

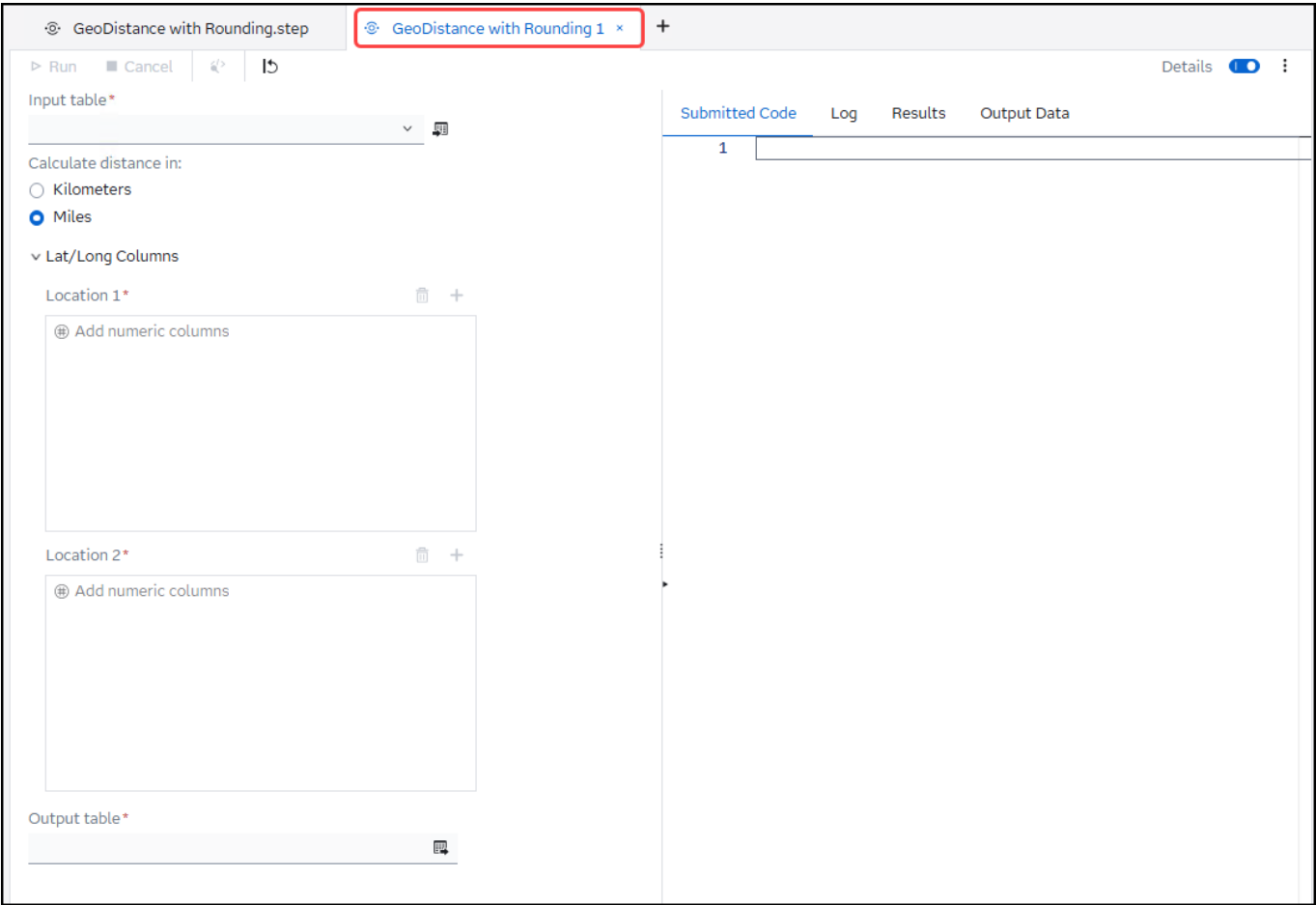


Setting visibility to *false* means that the user will not be able to view this control. The column name will be controlled by the code, but is needed for the output options in the port details you will set later in this exercise.

Add an **Output Table** *Data* control to the main canvas. Enter the properties as depicted below.



Save the Custom Step. Click **Launch** to preview the design of Page 1 - *Calculate Geo Distance*.

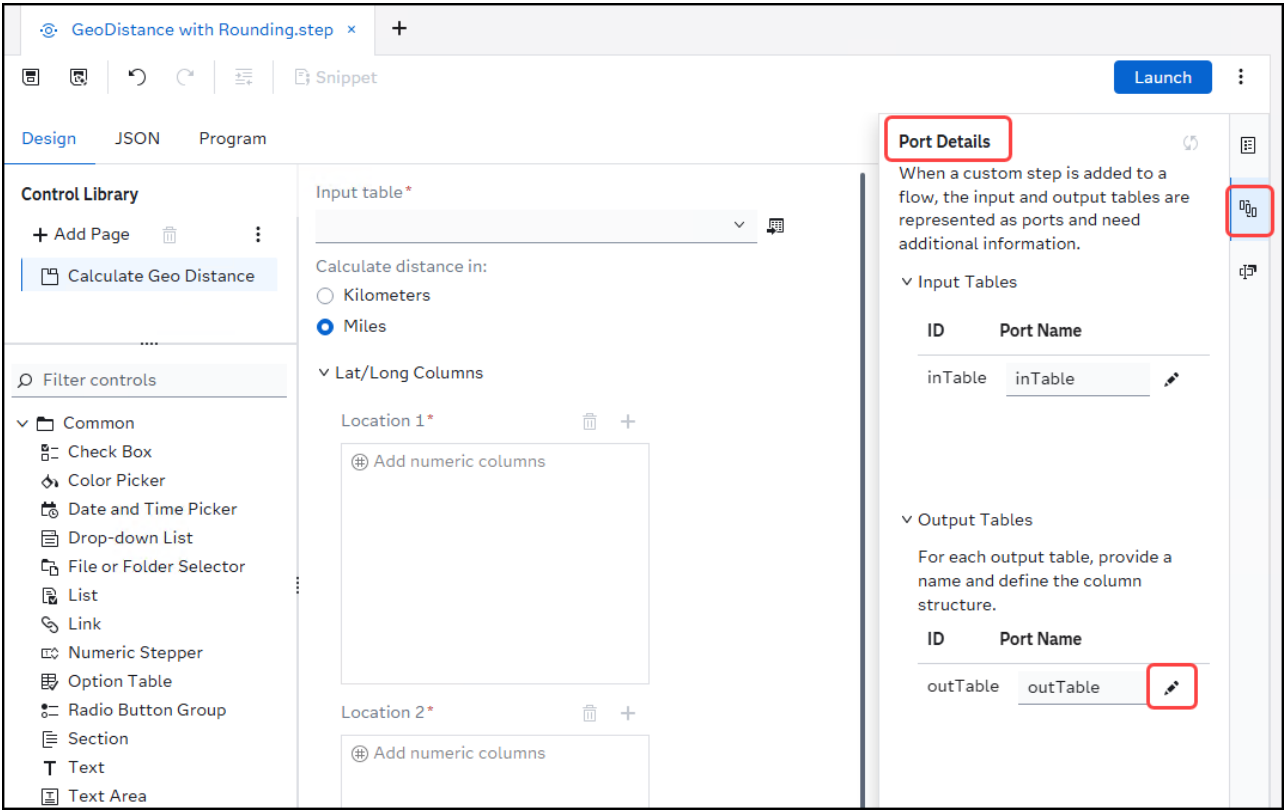


 The *new column* is not displayed in the Preview since its visibility is set to *false*.

Close the launched custom step when done.

Set Port Details

- 1. Select  to enter the port details for the custom step.
- 2. Select  for the **outTable** *Output Tables* port to edit its details.



3. Select **All columns** → **Input table (inTable)** to add all the columns from that input table to the output.

×

Output Table Metadata

▼ Details

ID:
outTable

Port name: *
outTable

Description:

▼ Define the column structure of the output table ?

Drop columns ▼ Keep columns ▼ New column ▼

All columns ▼
Input table (inTable)

No items added.

OK Cancel

4. Select **New column** → **Distance between the locations (newColumn)** to add that new column to the output.

×

Output Table Metadata

▼ Details

ID:
outTable

Port name: *

outTable

Description:

▼ Define the column structure of the output table

?

Drop columns ▼

Keep columns ▼

New column ▼

All columns ▼

Data Controls	Linked Input Control	Publish
↗ Input table ID: inTable		All
⌕+ Distance between the locations ID: newColumn		New

OK

Cancel

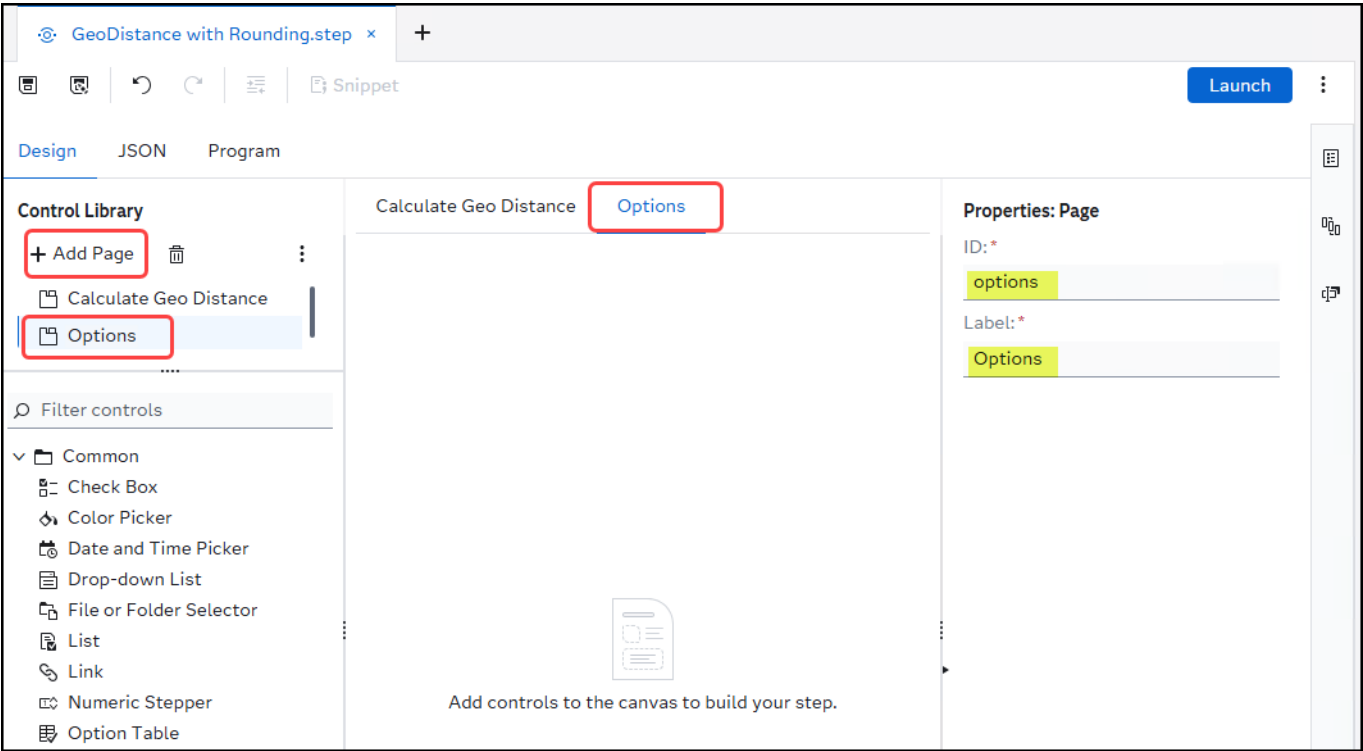
5. Click **OK** to save the changes to output table metadata.
6. **Save** the changes to the custom step.

Assignment: Build Page 2 UI

Create a second page for the custom step by selecting

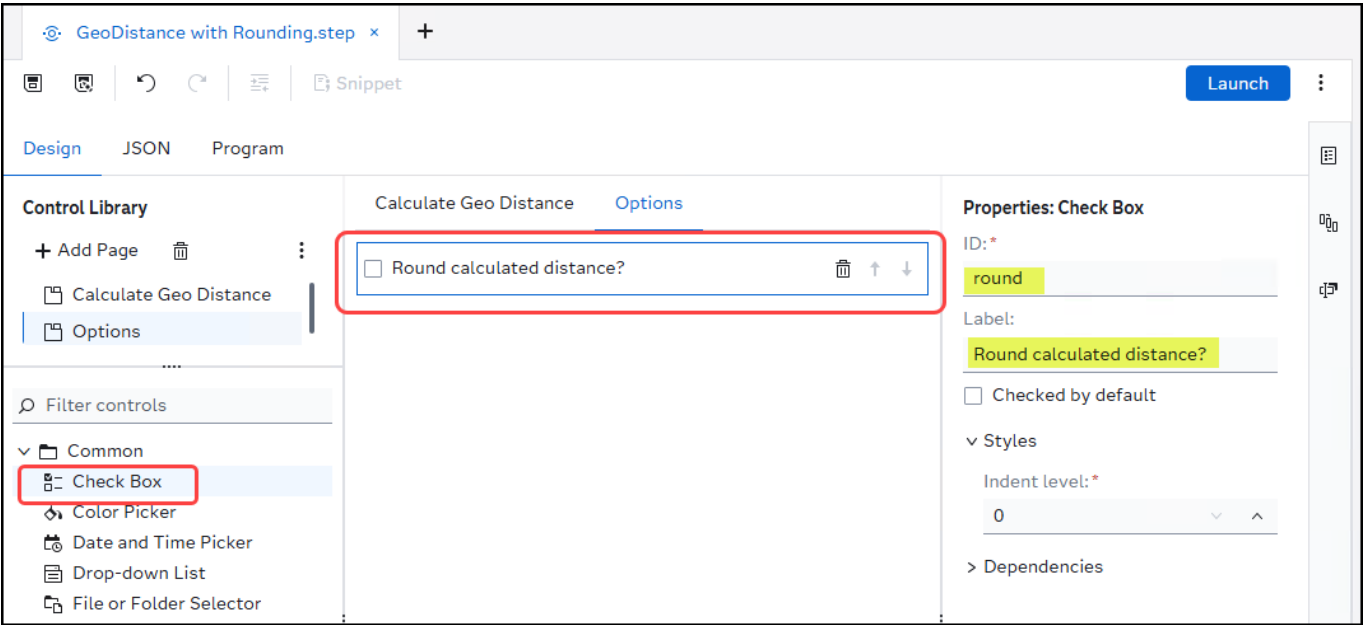
+ Add Page

 in the *Control Library* on the *Designer* tab. Enter the properties as depicted below.



Save the custom step.

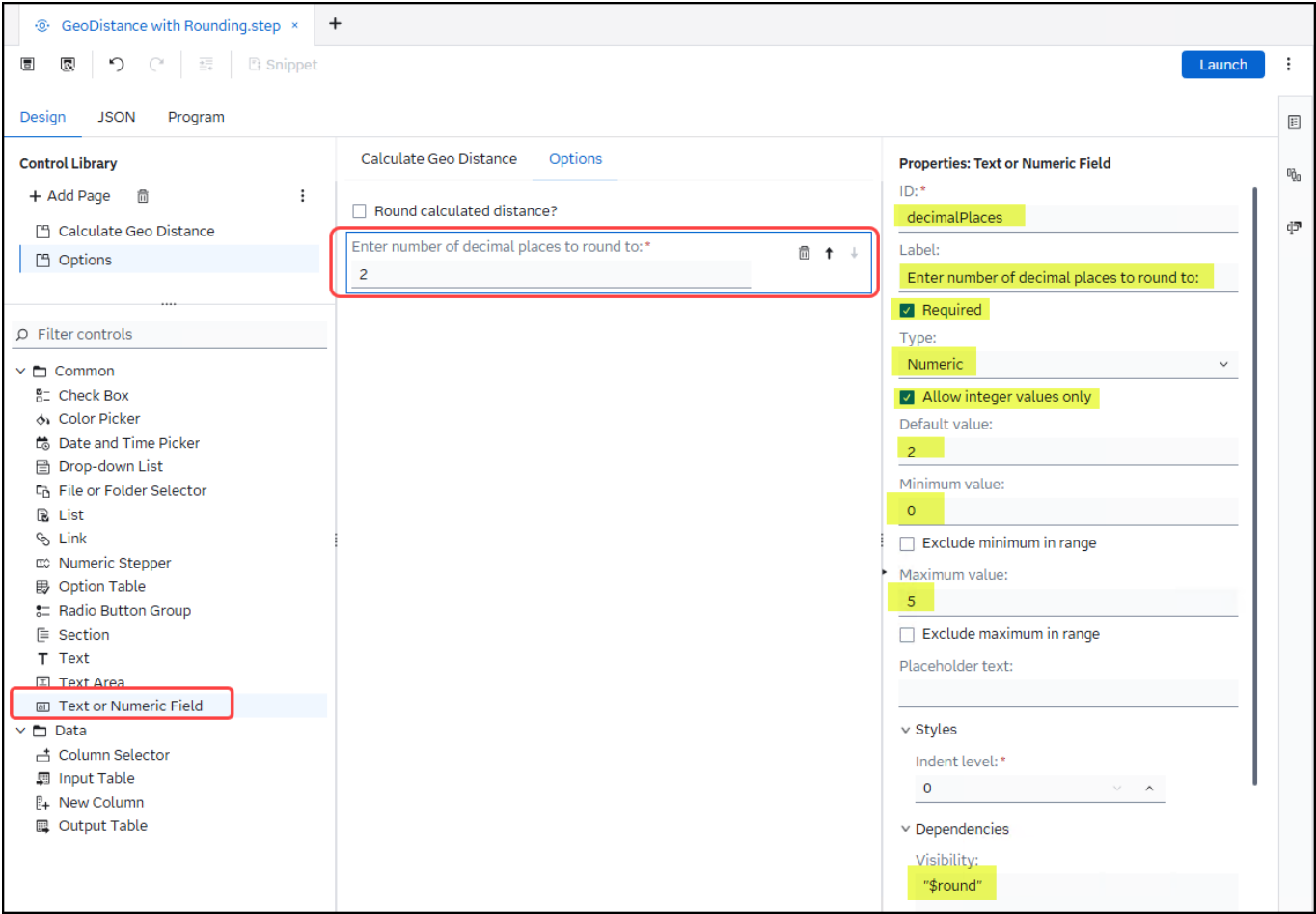
Add a **Check Box** *Common* control to the canvas on *Page2*. Enter the properties as depicted below.



Save the custom step.

Add a **Text or Numeric Field** *Common* control to the canvas. Enter the properties as depicted below.

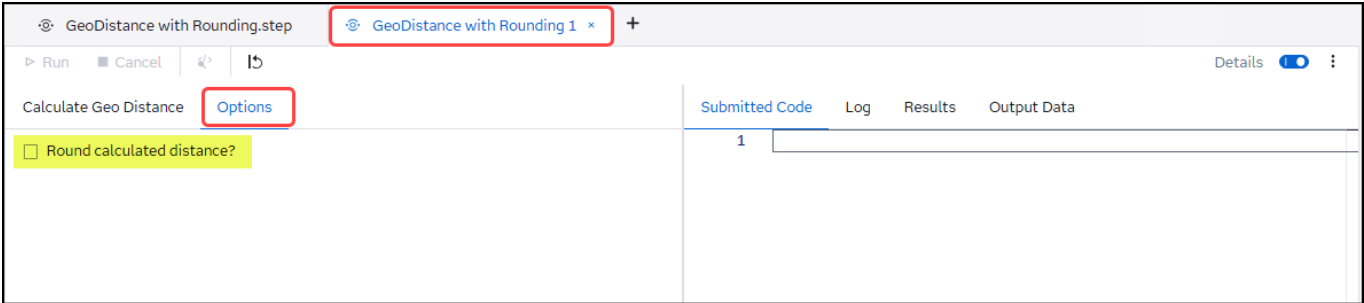
✎ "\$round" in the *Dependencies Visibility* section indicates that this control is dependent on the **round** Check Box control and will only be displayed if it is checked.



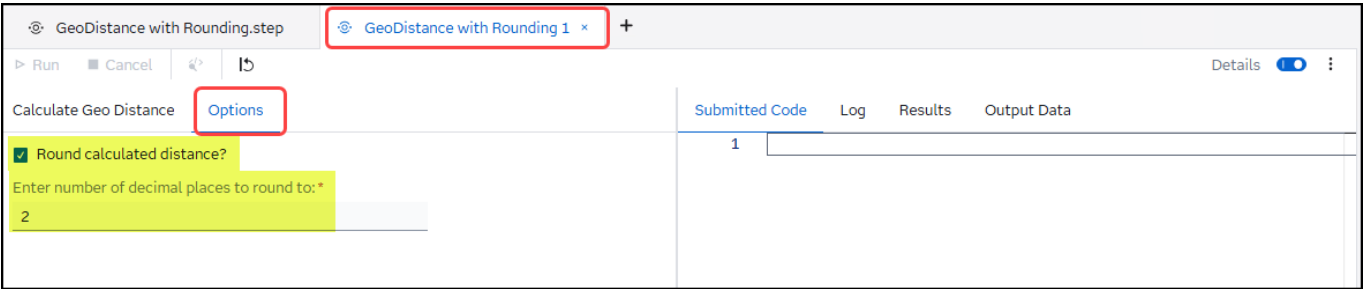
Save the custom step. Select **Launch** to preview Page 2 - *Options* of the custom step.

Confirm that the dependent **decimalPlaces** selection is only displayed if the **round** check box is checked.

Round Option Unchecked



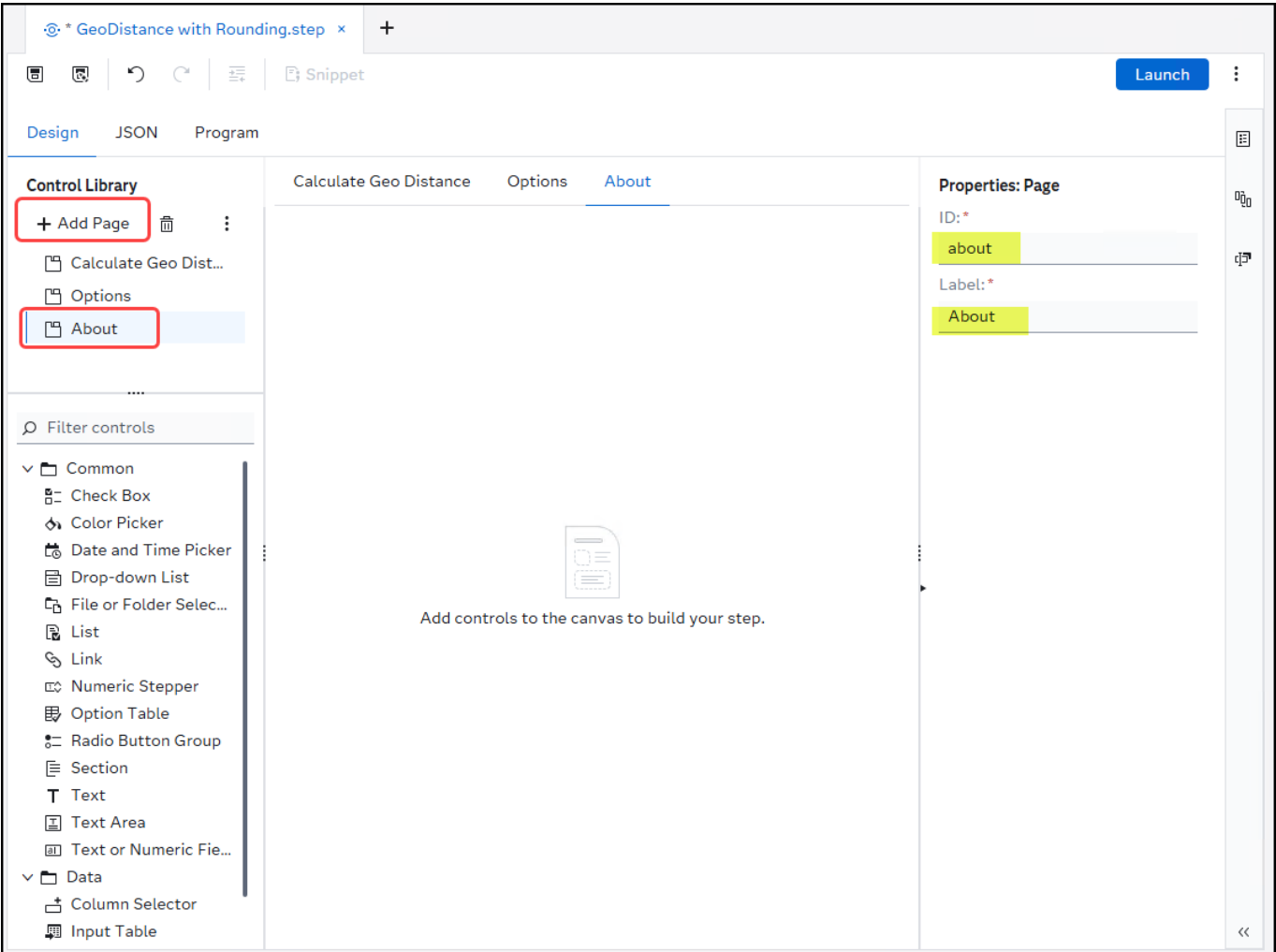
Round Option Checked



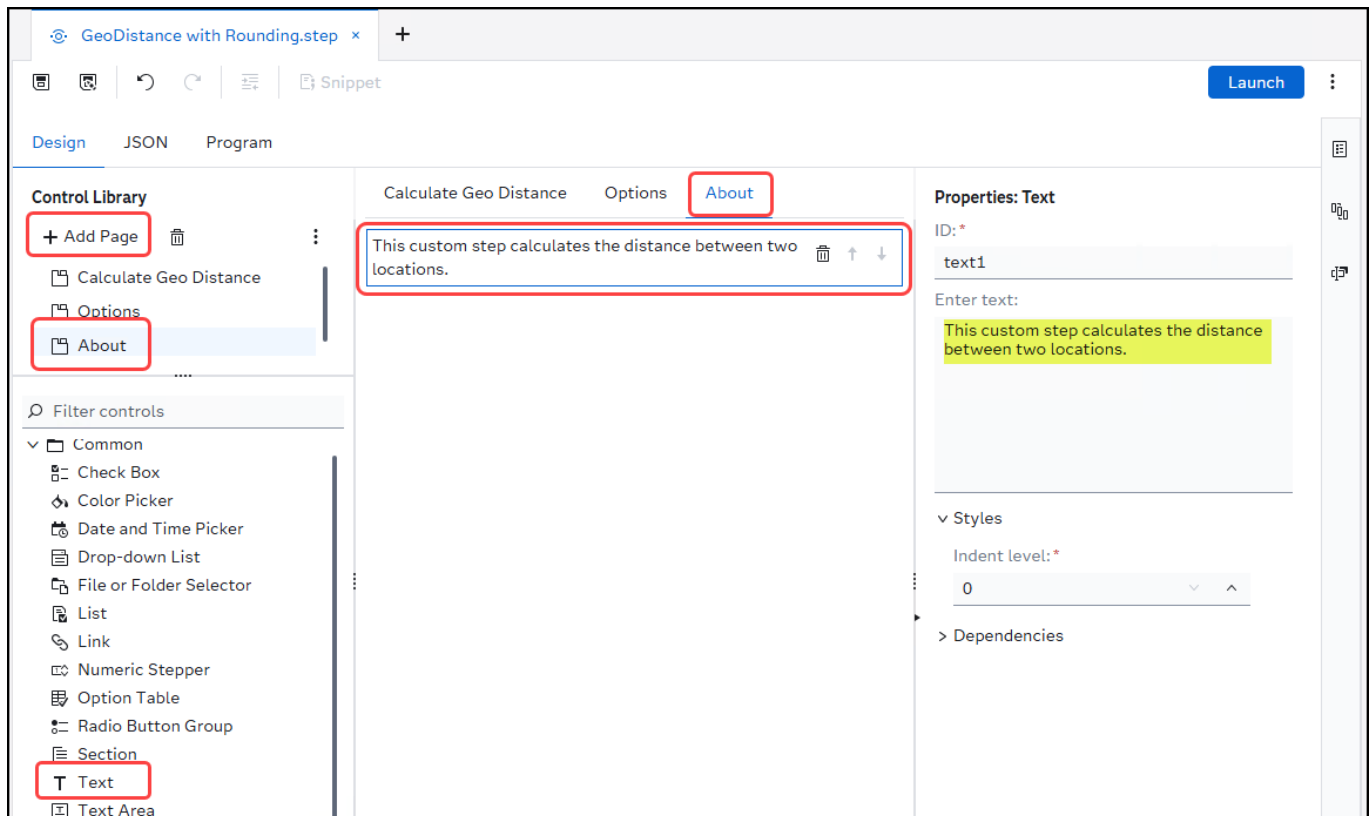
Close the launched custom step when done.

Assignment: Build Page 3 UI

Add an **About** page (tab) as depicted below.



On the *About* tab, use the **Text** control in the *Common* section to add the following text: **This custom step calculates the distance between two locations.**



Save the custom step.

Add Code to the Custom Step

1. Select the **Program** tab on the custom step.
2. Copy and Paste the following code:

```
/* Set Input and Output Data */
data &outTable ;
set &inTable ;

/* newColumn is a hidden control in the UI. The code sets the name of the new
output column. */

/* Calculate GeoDistance in Miles */
%if &disType=Miles %then %do ;
    %let newColumn_name=distance_in_Miles ;
    &newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name,
&loc2_2_name, 'M') ;
%end;

/* Calculate GeoDistance in Kilometers */
%else %do ;
    %let newColumn_name=distance_in_Kilometers ;
    &newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name,
&loc2_2_name, 'K') ;
```

```

%end ;

/* Round GeoDistance Value */
%if &round=1 %then %do ;
    select(&decimalPlaces) ;
        when(0) &newColumn_name=round(distance_in_&disType) ;
        when(1) &newColumn_name=round(distance_in_&disType,0.1) ;
        when(2) &newColumn_name=round(distance_in_&disType,0.01) ;
        when(3) &newColumn_name=round(distance_in_&disType,0.001) ;
        when(4) &newColumn_name=round(distance_in_&disType,0.0001) ;
        when(5) &newColumn_name=round(distance_in_&disType,0.00001) ;
    end;
%end;
run;

```

!! The **ID** names for the controls are used in the code for the custom step, so ensure you entered them correctly; otherwise, you will either need to edit the code or the control ID in the properties on the *Design* tab.

```

1  /* Set Input and Output Data */
2  data &outTable ;
3      set &inTable ;
4
5
6  /* newColumn is a hidden control in the UI. The code sets the name of the new output column. */
7
8  /* Calculate GeoDistance in Miles */
9  %if &disType=Miles %then %do ;
10     %let newColumn_name=distance_in_Miles ;
11     &newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name, &loc2_2_name, 'M') ;
12 %end;
13
14 /* Calculate GeoDistance in Kilometers */
15 %else %do ;
16     %let newColumn_name=distance_in_Kilometers ;
17     &newColumn_name=geodist(&loc1_1_name, &loc1_2_name, &loc2_1_name, &loc2_2_name, 'K') ;
18 %end ;
19
20
21 /* Round GeoDistance Value */
22 %if &round=1 %then %do ;
23     select(&decimalPlaces) ;
24         when(0) &newColumn_name=round(distance_in_&disType) ;
25         when(1) &newColumn_name=round(distance_in_&disType,0.1) ;
26         when(2) &newColumn_name=round(distance_in_&disType,0.01) ;
27         when(3) &newColumn_name=round(distance_in_&disType,0.001) ;
28         when(4) &newColumn_name=round(distance_in_&disType,0.0001) ;
29         when(5) &newColumn_name=round(distance_in_&disType,0.00001) ;
30     end;
31 %end;
32 run;

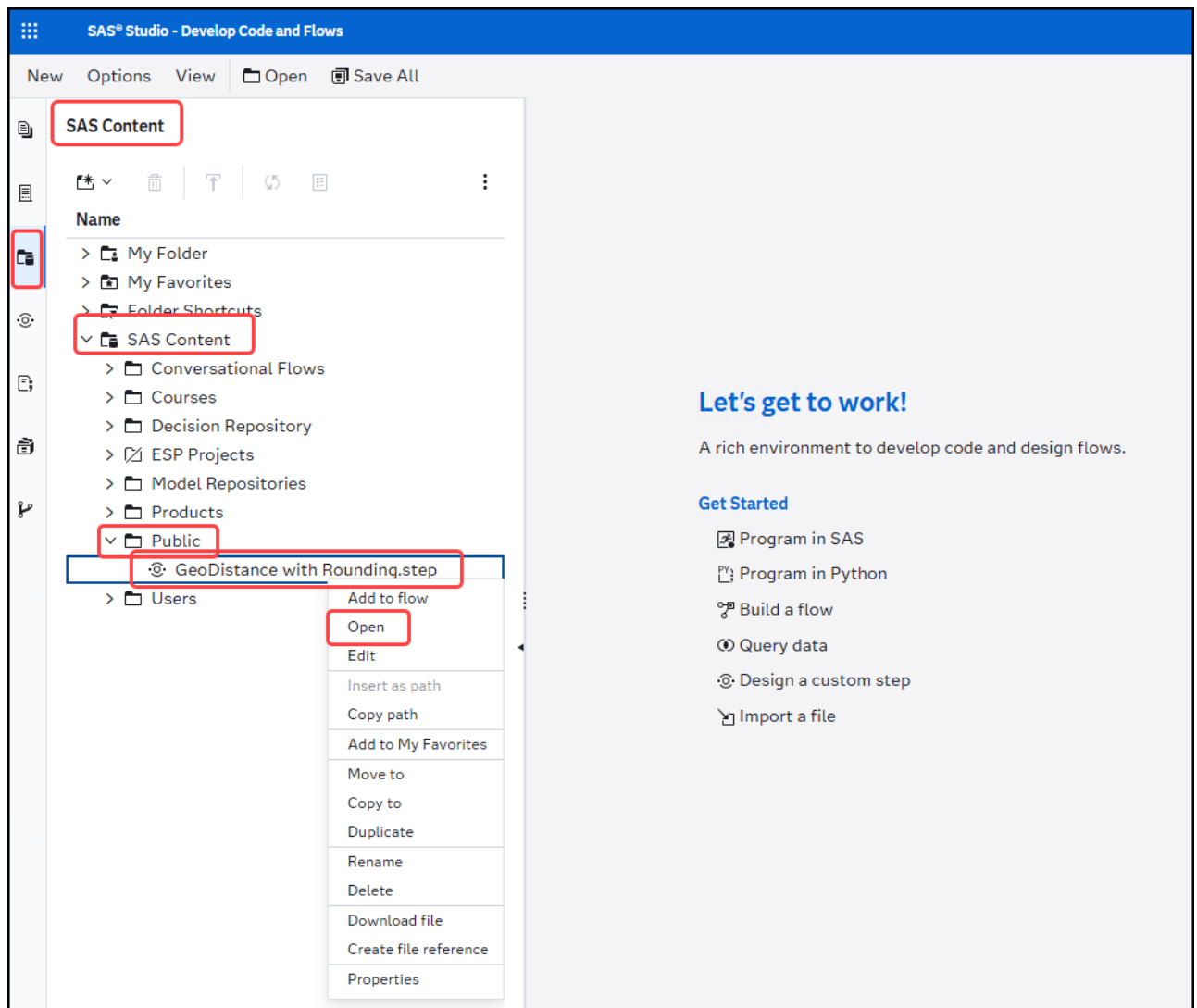
```

3. **Save** and close the custom step.

Open and Test the Custom Step in Stand-Alone Mode



1. Select  to view the **SAS Content** tab in *SAS Studio*.
2. Navigate to **SAS Content → Public**.
3. Select **GeoDistance with Rounding.step** and *right-click* to view its menu.
4. Select **Open** to view the custom step in stand-alone mode.



Assignment: Test the Custom Step



Perform a test using **MAPS.AFRICA** as the input table. Use  to select the input table.

Select columns **X** and **Y** for *Location 1* and columns **LAT** and **LONG** for *Location 2*.



Select **WORK.TEST** as the output table. Use  to select the output table.

 You will need to type the table name of **TEST** since it does not already exist in the **WORK** library.

Run the custom step and review the *Output Data* tab.

GeoDistance with Rounding

Run Cancel

Calculate Geo Distance Options About

MAPS.AFRICA

Calculate distance in:
☐ Kilometers
☒ Miles

Lat/Long Columns

Location 1*
☐ X
☐ Y

Location 2*
☐ LAT
☐ LONG

Output table*
WORK.TEST

Submitted Code Log Results **Output Data (1)**

WORK.TEST Columns: 9 Rows: 52,824

Enter expression

#	LAT	LONG	distance_in_Miles
1	0.4763100675	0.1512618584	65.472588797
2	0.4665992501	0.1342788354	63.89790514
3	0.4560400078	0.1162873997	62.217721563
4	0.4472357814	0.1016120945	60.839560605
5	0.4363323078	0.083874638	59.162371164
6	0.4288467945	0.0719608896	58.028930644
7	0.4214776259	0.0604368686	56.927055882
8	0.4099293542	0.0427314778	55.225587043
9	0.4018329513	0.0305529678	54.050067918
10	0.3946674438	0.01994523	53.021995608
11	0.3866631509	0.0082563867	51.886056752
12	0.3810150668	0.0001405863	51.093599576
13	0.3683080954	-0.020415509	49.240100017
14	0.3675275593	-0.020691838	49.162707119
15	0.3665627803	-0.020895479	49.072022632
16	0.366048863	-0.020725785	49.033786438
17	0.3654283163	-0.020813046	48.977050129
18	0.3647834953	-0.020706392	48.925261187
19	0.3631496969	-0.02025552	48.80073741

View the resulting **TEST** table in the **WORK** library. Select  to view the library connections.

SAS Studio - Develop Code and Flows

New Options View Open Save All

Library Connections

Connected Save

Search (library;table;...)

Name

- MAPS
- MAPSGFK
- MAPSSAS
- SASDQREF
- SASHELP
- SASUSER
- WORK
 - TEST

GeoDistance with Rounding

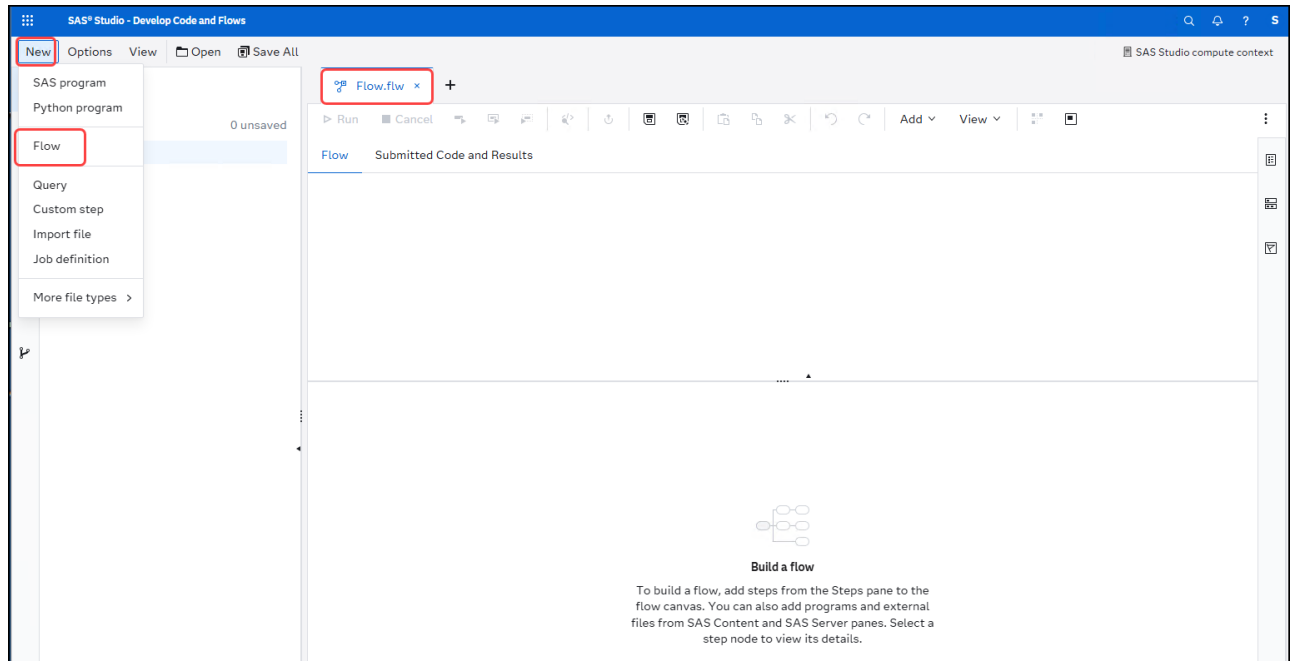
WORK.TEST Columns: 9 Rows: 52,824

Enter expression

#	CONT	ID	SEGMENT	DENSITY	X	Y	LAT	LONG	distance_in_Miles
1	94	125	1	0	-0.454145736	0.355554876	0.4763100675	0.1512618584	65.472588797
2	94	125	1	5	-0.43878874	0.3453814464	0.4665992501	0.1342788354	63.89790514
3	94	125	1	3	-0.422490668	0.3343443053	0.4560400078	0.1162873997	62.217721563
4	94	125	1	5	-0.409172181	0.3251576359	0.4472357814	0.1016120945	60.839560605
5	94	125	1	0	-0.393045542	0.3138009477	0.4363323078	0.083874638	59.162371164
6	94	125	1	6	-0.382194958	0.3060158501	0.4288467945	0.0719608896	58.028930644
7	94	125	1	5	-0.37168453	0.2983602571	0.4214776259	0.0604368686	56.927055882
8	94	125	1	3	-0.355505795	0.286376767	0.4099293542	0.0427314778	55.225587043
9	94	125	1	6	-0.344354685	0.2779837276	0.4018329513	0.0305529678	54.050067918
10	94	125	1	5	-0.33462672	0.2705617363	0.3946674438	0.01994523	53.021995608
11	94	125	1	6	-0.323889858	0.262276229	0.3866631509	0.0082563867	51.886056752
12	94	125	1	3	-0.316424986	0.2564342568	0.3810150668	0.0001405863	51.093599576
13	94	125	1	1	-0.29740248	0.2431999799	0.3683080954	-0.020415509	49.240100017
14	94	125	1	6	-0.297170436	0.2424252066	0.3675275593	-0.020691838	49.162707119
15	94	125	1	3	-0.297014807	0.2414726527	0.3665627803	-0.020895479	49.072022632

Close the *TEST* table and the custom step when done.

1. In **SAS Studio**, select **New → Flow** from the menu to create a new flow file.

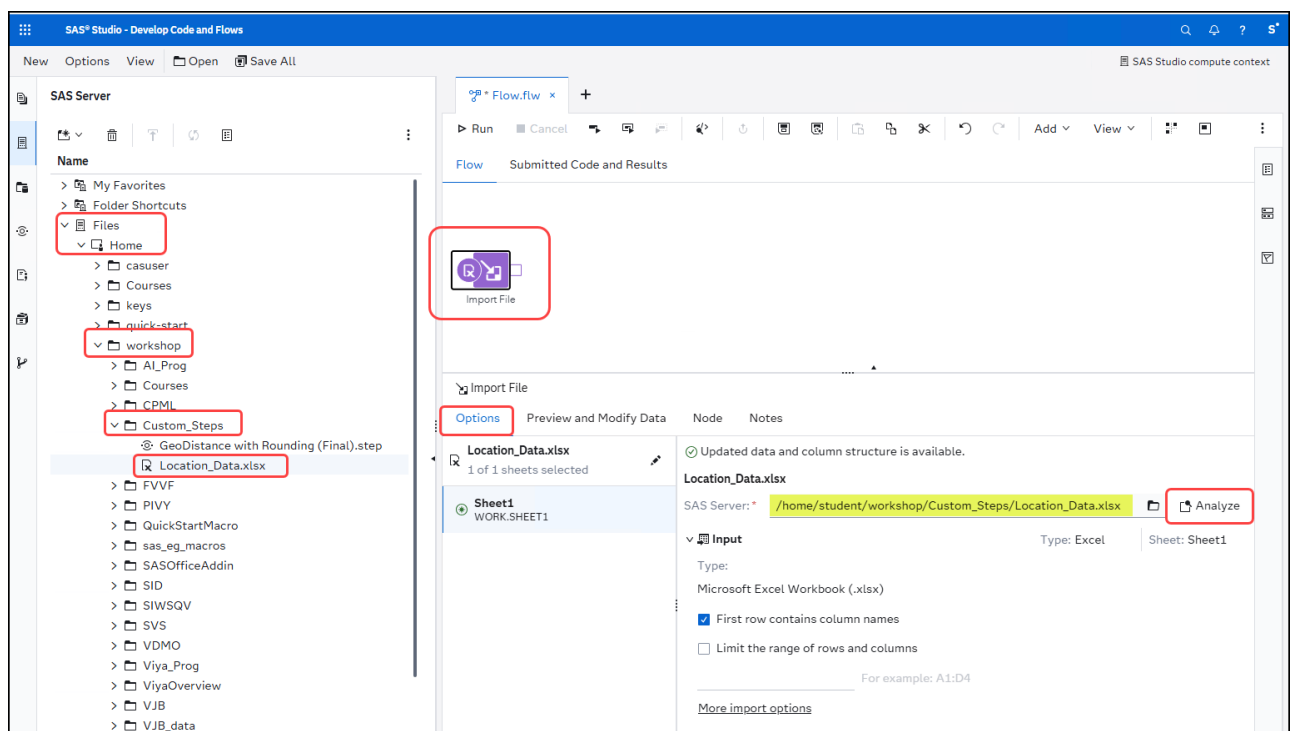


2. Select  to view the files on the SAS Server.

3. Expand the folders **Files → Home → workshop → Custom_Steps**.

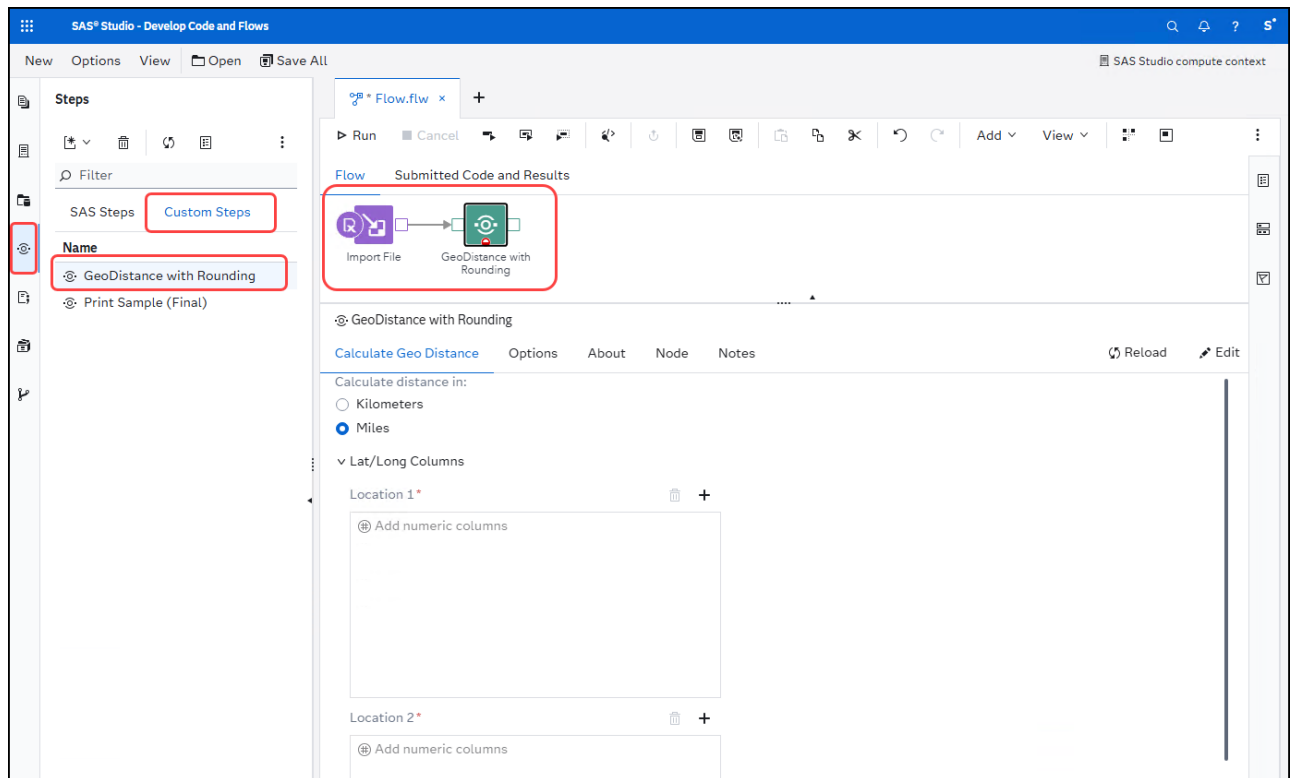
4. Select the file **Location_Data.xlsx** and drag it on to the Flow canvas to add an *Import File* step for that spreadsheet.

5. Click **Analyze** on the step *Options* tab to analyze and add the column names and contents of the file.



6. Select  to view the Steps pane.

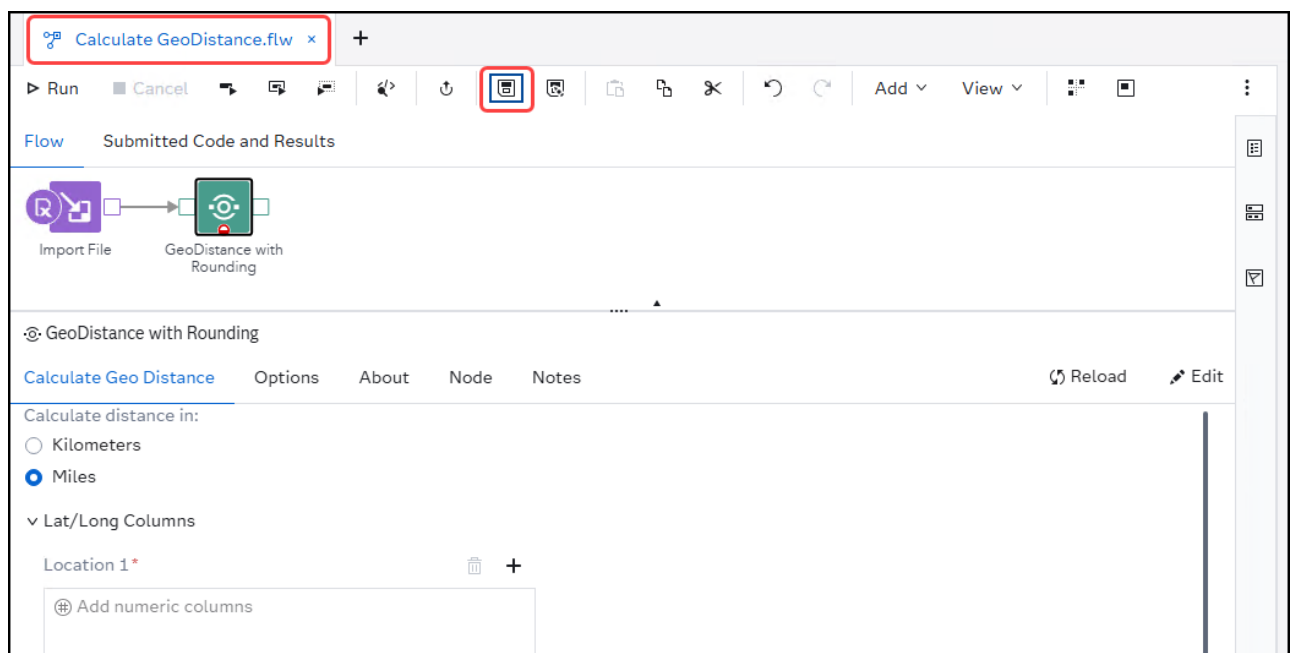
7. Select the **Custom Steps** tab to view the available custom steps.
8. Drag the **GeoDistance with Rounding** custom step to the flow canvas and use the mouse pointer to connect it to the *Import File* step.



9. Click  to save the flow file.

10. Navigate to **SAS Content** → **Public**.

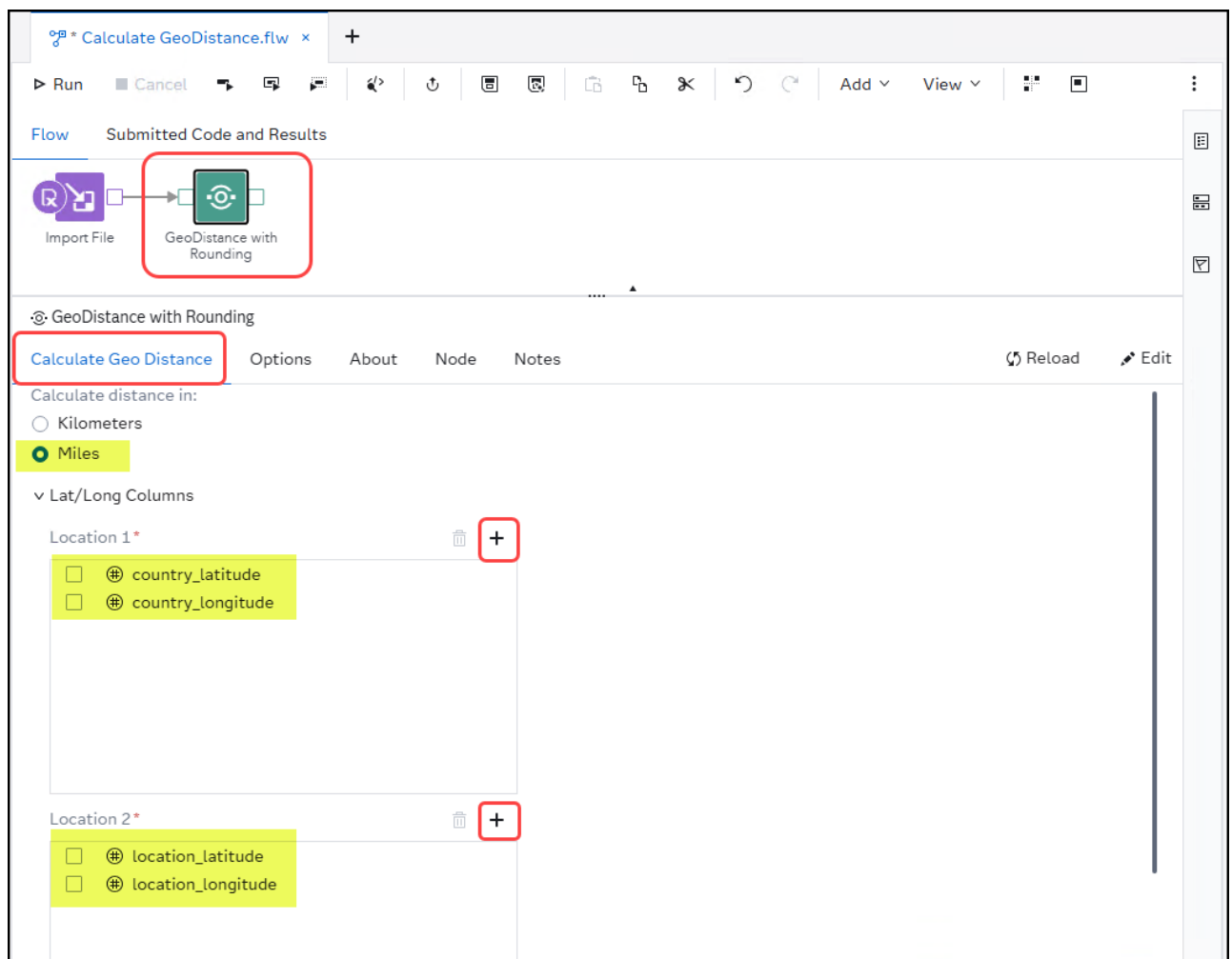
11. Enter **Calculate GeoDistance** for the *name* and click **Save**.



12. Select the **GeoDistance with Rounding** step in the flow canvas to view its properties in the bottom pane.

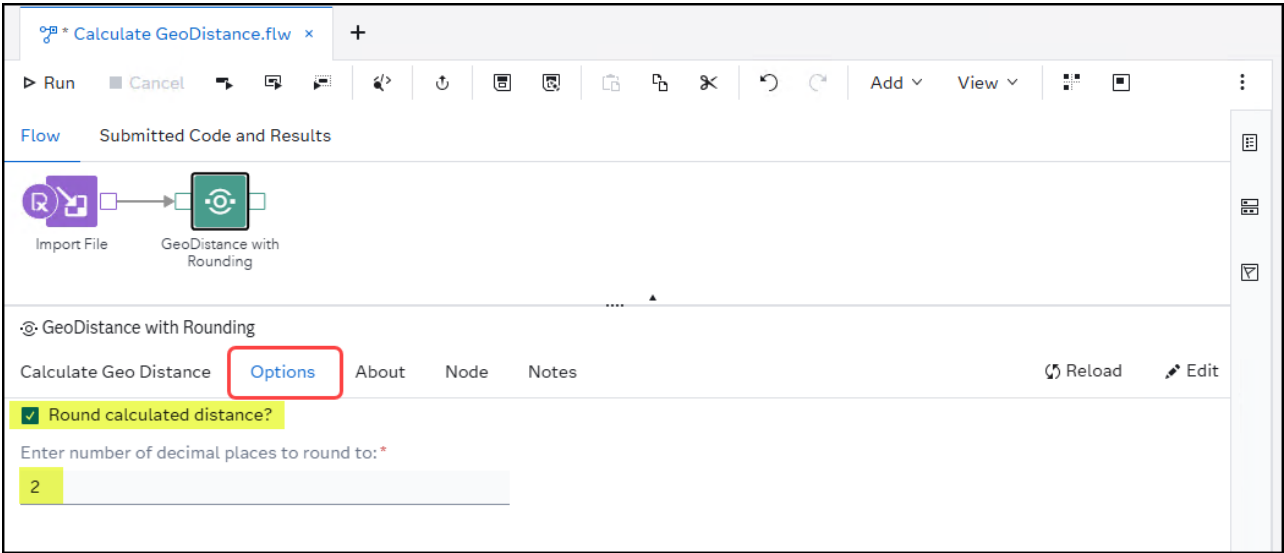
13. On the *Calculate Geo Distance** tab, enter the following:

- Calculate distance in:
 - ☐ Kilometers
 - ☒ Miles
- Location 1
 - **country_latitude**
 - **country_longitude**
- Location 2
 - **location_latitude**
 - **location_longitude**

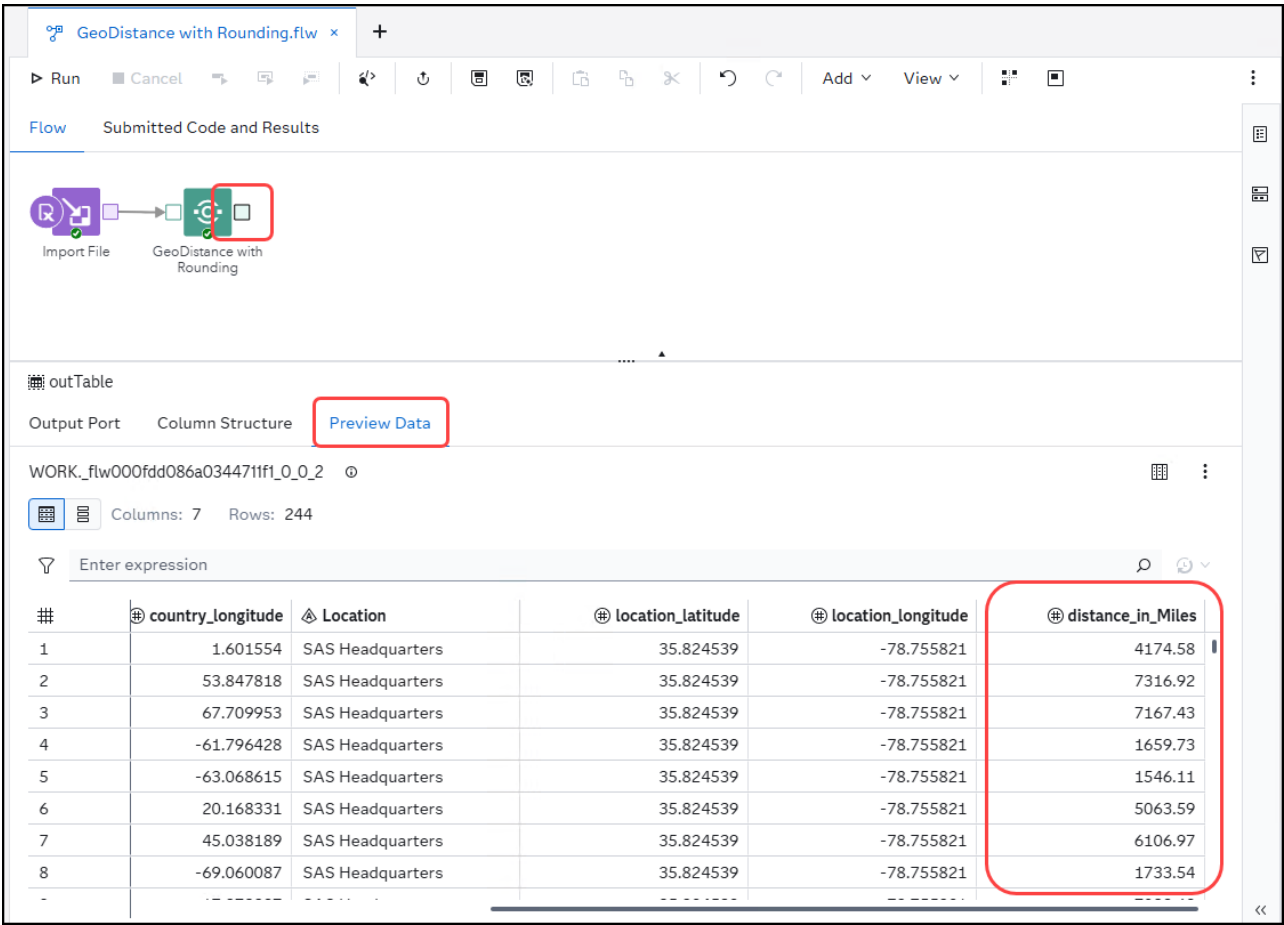


14. Select the **Options** tab for the custom step properties and enter the following:

- ☒ Round calculated distance?
- Enter number of decimal places to round to:
2



- 15. **Save** the flow file.
- 16. Click **Run** to execute the flow file.
- 17. Select the **output port** for the *GeoDistance with Rounding* step to view the results.
- 18. Select the **Preview Data** tab.

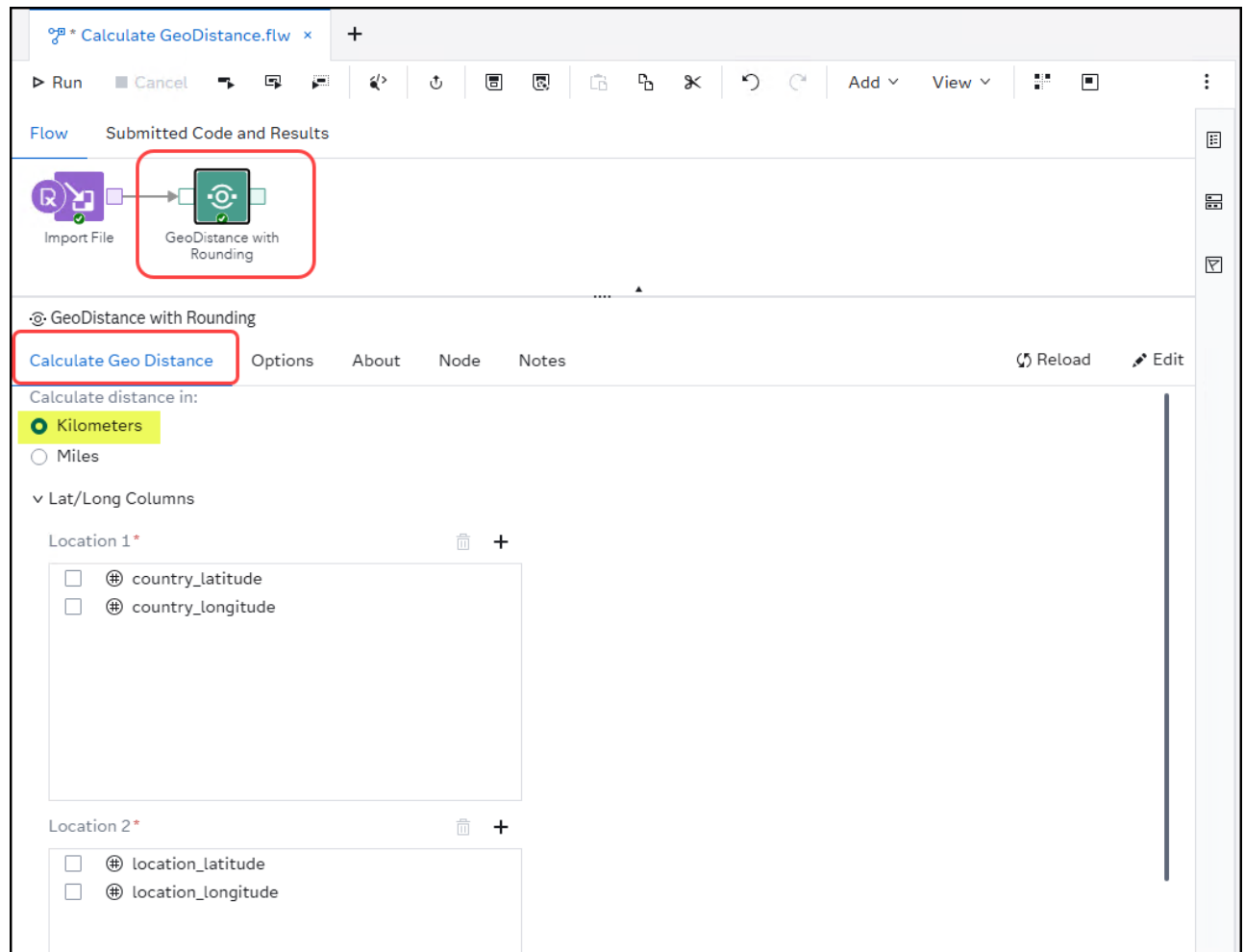


- 19. Select the **GeoDistance with Rounding** step in the flow canvas to view its properties in the bottom pane.
- 20. On the **Calculate Geo Distance** tab, change the following selection:

- Calculate distance in:

☒ Kilometers

☐ Miles



21. **Save** the flow file.

22. Click **Run** to execute the flow file.

23. Select the **output port** for the *GeoDistance with Rounding* step to view the results.

24. Select the **Preview Data** tab.

Calculate GeoDistance.flow

Run Cancel

Flow Submitted Code and Results

Import File GeoDistance with Rounding

Output Port Column Structure Preview Data

WORK_flw000bfd8cf3013ed11f1_0_0_2

Columns: 7 Rows: 244

Enter expression

#	longitude	Location	location_latitude	location_longitude	distance_in_Kilometers
1	1.601554	SAS Headquarters	35.824539	-78.755821	6718.34
2	3.847818	SAS Headquarters	35.824539	-78.755821	11775.44
3	7.709953	SAS Headquarters	35.824539	-78.755821	11534.85
4	1.796428	SAS Headquarters	35.824539	-78.755821	2671.08
5	3.068615	SAS Headquarters	35.824539	-78.755821	2488.23
6	0.168331	SAS Headquarters	35.824539	-78.755821	8149.05
7	5.038189	SAS Headquarters	35.824539	-78.755821	9828.22
8	9.060087	SAS Headquarters	35.824539	-78.755821	2789.86
9	7.873887	SAS Headquarters	35.824539	-78.755821	11327.66
10	0.071389	SAS Headquarters	35.824539	-78.755821	13509.15
11	3.616672	SAS Headquarters	35.824539	-78.755821	8366.32
12	0.132217	SAS Headquarters	35.824539	-78.755821	11050.09

25. Click **x** close the flow file.

Exercise Completed

You have completed the exercise for **Create Custom Steps with SAS Studio.**

THANK YOU FOR ATTENDING THIS SESSION!

PLEASE COMPLETE THE EVALUATION!

Custom Steps

Resources



Training Course

- [Developing Custom Steps with SAS® Studio Analyst](#)

Documentation

- [SAS Help Center: SAS Studio: Working with Custom Steps](#)

SAS Community Articles

- [Search - SAS Support Communities for Custom Steps](#)

Quick Start Video

- [Create Custom Steps with SAS Studio](#)

YouTube Video Series

- [Creating a Custom Step in SAS Studio](#)

GitHub Repository of Custom Steps

- <https://github.com/sassoftware/sas-studio-custom-steps>

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Training Course:

- [Developing Custom Steps with SAS® Studio Analyst:](#)
<https://learn.sas.com/course/view.php?id=220>

Documentation:

- [SAS Help Center: SAS Studio: Working with Custom Steps:](#)
<https://go.documentation.sas.com/doc/en/sasstudiocdc/default/webeditorcdc/webeditorsteps/titlepage.htm>

SAS Community Articles:

- [Search - SAS Support Communities for Custom Steps:](#)
https://communities.sas.com/t5/forums/searchpage/tab/message?q=Custom%20Steps&noSynonym=false&collapse_discussion=true

Quick Start Video:

- [Create Custom Steps with SAS Studio:](#)
<https://video.sas.com/detail/video/6347011003112/create-custom-steps-with-sas-studio>

YouTube Video Series:

- [Creating a Custom Step in SAS Studio:](#)
<https://youtube.com/playlist?list=PLVV6eZFA22QyLhfnKKW3k08XVx1KOZCJ4&si=htu2O3BmS05oo9aD>

GitHub Repository of Custom Steps:

- <https://github.com/sassoftware/sas-studio-custom-steps>

Thanks for attending this Learning Lab session!

Please complete the evaluation for the session.

Mary Kathryn Queen
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